

School of Computer Science

FACULTY OF ENGINEERING AND
PHYSICAL SCIENCES



UNIVERSITY OF LEEDS

Advanced techniques for predicting left shifts in patient flow using predictive process monitoring

Fenil Kalasariya

**Submitted in accordance with the requirements for the degree of
MSc Advanced Computer Science (Data Analytics)**

2025/2026

The candidate confirms that the following have been submitted.

Items	Format	Recipient(s) and Date
Deliverable 1, 3, 5	Report	SSO (31/08/26)
Deliverable 2, 4	URL: https://github.com/Fenilll21/msc-dissertation	SSO (31/08/26)

Type of project: Empirical Investigation

The candidate confirms that the work submitted is their own and the appropriate credit has been given where reference has been made to the work of others.

I understand that failure to attribute material which is obtained from another source may be considered as plagiarism.

(Signature of Student)



Summary

Hospitals are constrained by how patients move through the system rather than by how many arrive. A left shift is an unplanned move to a higher level of care within the same hospital admission. This project predicts left shifts on MIMIC-IV using outcome-oriented predictive process monitoring, extending the prior work of Aljebreen et al. [1]. An event log was built from the *transfers* table and two cohorts (ICU and SW) were defined, each with its own prediction point. A total of thirteen models were compared on the same set of admissions and the same partition. Logistic regression, random forest, gradient boosting, k -nearest neighbours and a multilayer perceptron were built over the tabular features and RNN, LSTM, GRU, BiLSTM, BiGRU, two CNN variants and a CNN-LSTM hybrid were built over the event sequences. The pipeline was then re-run under five label variants and on a restricted cohort, so that only the outcome rule changed.

The outcome definition moved performance further than the model did. Gradient boosting was the strongest model in every cohort and label variant, and the sequence models were indistinguishable from one another once the tabular features were present, as the traces were too short for the sequence structure to help. Most of the apparent gap between the two cohorts disappeared once admissions that cannot produce a positive outcome were removed. A care-intensity mapping built from the Intensive Care Society levels classifies most ICU left shifts as re-escalations but few SW ones, so the two labels may not name the same clinical event. The ICU baseline reproduced closely, the SW did not. No label was validated by clinical adjudication, so the results measure agreement with the rule rather than with the deterioration that the rule names.

Acknowledgements

First, I would like to express my gratitude to my supervisor Dr. Owen Johnson, for his continuous guidance, support, and encouragement throughout the entire project. His insightful recommendations consistently pushed the work towards what the data could actually support, and the direction the dissertation took owes a great deal to those conversations.

I am equally grateful to my assessor, Dr. Brandon Bennett, for his feedback on the project. I am also grateful to Abdulaziz Aljebreen, whose work this project extends, for answering my questions about his definitions and methods and for his help in replicating his results.

Finally, I thank my family, Mom, Dad and my little brother Vraj for their support throughout the year, and my friends Dhruv, Prachita, Dev, Aditya, Shukan, Tanuj and Pranshav, who kept me going through the long stretches of this project.

Contents

Abbreviations	1
1 Introduction	2
1.1 Project Aim	2
1.1.1 Motivation	2
1.1.2 Process Mining	2
1.1.3 Left Shifts	3
1.1.4 Standard Techniques	4
1.2 Objectives	4
1.3 Deliverables	5
1.4 Ethical, legal, and social issues	5
1.4.1 Ethical	5
1.4.2 Legal	5
1.4.3 Social	6
2 Background Research	7
2.1 Literature Survey	7
2.1.1 Unplanned escalation as a clinical and operational problem	7
2.1.2 Process Mining and predictive process monitoring in healthcare	8
2.1.3 Prior work predicting escalation	10
2.1.4 What the reviews report about the field	12
2.1.5 Label quality and construct validity in clinical ML	13
2.1.6 Research gap	14
2.2 Methods and Techniques	16
2.2.1 Event log construction and activity abstraction	16
2.2.2 Pipeline design decisions	17
2.2.3 Choice of classifier	18
2.2.4 Analysis of fitted models	18
2.2.5 Evaluation design	19
2.2.6 Defining and validating the outcome	19
2.3 Choice of methods	20
3 Datasets and Experimental Design	23
3.1 Datasets and data sources	23
3.1.1 Data source, access and extraction	23

3.1.2	Event log construction	24
3.1.3	Activity abstraction	25
3.1.4	Cohorts and prediction points	26
3.1.5	The dependent variable	28
3.1.6	Independent variables	30
3.1.7	Control-flow representations	31
3.1.8	Care-intensity mapping	32
3.2	Experimental design	33
3.2.1	Prediction framework and prefix construction	33
3.2.2	Data partitioning	33
3.2.3	Feature conditions	34
3.2.4	Model families, configuration and repetition	35
3.2.5	Metrics and operating point	36
3.2.6	Attribution and error analysis	36
3.2.7	Robustness designs: label sensitivity and restricted cohort	37
3.2.8	Computational environment and reproducibility	38
4	Results of the Empirical Investigation	39
4.1	Experimental programme and reporting conventions	39
4.2	Replication of the published baseline	40
4.3	Model families and the value of process features	41
4.4	Sensitivity to outcome definition	43
4.5	Feature attribution and temporal validity	44
4.6	Error analysis	46
4.7	Care-intensity levels	48
5	Validation of Results	49
5.1	Validation strategy	49
5.2	Data-level validation	50
5.3	Partition integrity and the provenance gate	50
5.4	Reproduction of published results	52
5.5	Construct validity of the outcome definition	53
5.6	Robustness of the findings	54
5.7	Threats to validity	56
6	Conclusions and Future Work	58
6.1	Conclusions	58
6.2	Future work	59
	References	61

Appendices	64
A External Material	65
A.1 Systematic reviews of ICU readmission prediction	65
B Care-unit mappings	66
B.1 Activity grouping	66
B.2 Care units outside the mapping	66
C Pipeline reference	67
C.1 Source tables	67
C.2 Process model, dominant pathway	67
C.3 Cohort process models	69
C.4 The outcome as a process model	71
C.5 Control-flow representations	72
C.6 Software environments	73
C.7 Surgical ward baseline replication	73
C.8 Single-group and incremental ablation	74
D Ethical Issues Addressed	75
D.1 CITI Program Completion Report	75
D.2 Signed PhysioNet Data Use Agreement	77

List of Figures

1.1	Schematic representation of a generic hospital care pathway showing left shifts.	3
3.1	Event-log construction against the transfers table	24
3.2	Directly-follows model of the event log	26
3.3	Cohort construction	27
3.4	Three SW label variants on the feature matrix of Figure 3.3.	30
3.5	Intensive Care Society levels assigned to the 46 care units	32
3.6	Prediction point and prefix	33
4.1	Mean AUROC of the thirteen models under the <code>lat</code> label variant	41
4.2	Feature means in the FN cell against the TP cell	47
C.1	Dominant pathway of the event log	68
C.2	ICU cohort, frequency	69
C.3	SW cohort, frequency	69
C.4	ICU cohort, performance	70
C.5	SW cohort, performance	70
C.6	ICU cohort, <code>lat</code> positive class	71
C.7	ICU cohort, <code>lat</code> negative class	72

List of Tables

2.1	Design of the five primary studies	10
2.2	Stated outcome name against the operational rule	15
3.1	The five label variants and their positive rates	29
3.2	Canonical partitions	34
3.3	Feature conditions	34
3.4	Thirteen models, with number of fitted runs.	35
3.5	Restricted-cohort partitions	37
4.1	Baseline replication on <i>lat</i> label	40
4.2	Gradient Boosting against the best neural model and sequence model . . .	42
4.3	Best model per cohort and label variant	43
4.4	Leave-one-out ablation over the six feature groups with AUROC drops. .	45
4.5	Operating points for the error analysis	46
5.1	Partition digest with <i>split_id</i>	51
5.2	Agreement between care-intensity and ward-name label variants	54
5.3	Discharge outcome of structurally-negative group	55
A.1	Systematic reviews of ICU readmission prediction	65
B.1	Aggregation of the 46 careunit into five location groups	66
B.2	Care units outside the mapping	66
C.1	MIMIC-IV v3.1 tables used and the attributes each supplies (subsection 3.1.1). .	67
C.2	The 39 independent variables, by group	67
C.3	The two control-flow representations	72
C.4	Software environments	73
C.5	SW baseline replication	73
C.6	Single-group and incremental ablation	74

Abbreviations

Abbreviation	Expansion
AUC	Area under the curve
AUPRC	Area under the precision–recall curve
AUROC	Area under the receiver operating characteristic curve
BiGRU	Bidirectional gated recurrent unit
BiLSTM	Bidirectional long short-term memory
BMI	Body mass index
BoW	Bag of words
CI	Confidence interval
CNN	Convolutional neural network
ED	Emergency department
GB	Gradient boosting
GBM	Gradient boosting machine
GRU	Gated recurrent unit
HIPAA	Health Insurance Portability and Accountability Act
HPC	High-performance computing
ICD	International Classification of Diseases
ICS	Intensive Care Society
ICU	Intensive care unit
LSTM	Long short-term memory
MIMIC-III	Medical Information Mart for Intensive Care III
MIMIC-IV	Medical Information Mart for Intensive Care IV
ML	Machine learning
MLP	Multilayer perceptron
NHS	National Health Service
OOPPM	Outcome-oriented predictive process monitoring
PACU	Post-anaesthesia care unit
PPM	Predictive process monitoring
RF	Random forest
RNN	Recurrent neural network
SVM	Support vector machine
SW	Surgical ward

Chapter 1

Introduction

1.1 Project Aim

The aim of this project is to develop and use advanced techniques for predicting left shifts in patient flow. Hospitals are constrained by how patients move through the system and not how many arrive. This project extends the prior work of Aljebreen et al. [1] which applied Predictive Process Monitoring (PPM) to left-shift prediction on MIMIC-IV.

1.1.1 Motivation

NHS England reported that in Q4 2025/26 an average of 92.5% of available overnight general and acute beds occupied. This excludes critical care beds. [2] NHS operational planning guidance from 2020/21 expects the occupancy of general and acute beds to be a maximum of 92%. [3] The reported occupancy sits above the maximum guidance.

A delayed discharge is when a medically fit patient has not yet been discharged. Hospital delays are growing, and the growth is entirely coming from long-stay patients. Delayed patients in hospitals for 20 days or less are 5,992 per day, whilst those who are delayed for 21 days or longer rose about 8% from 5,959 to 6,413 between January 2022 and January 2026. The Nuffield Trust states that delayed discharges lead to delays to admission from Accident and Emergency (A&E), which worsens ambulance handover and response times. This creates a flow problem rather than a demand problem [4].

The UK recorded 2.4 beds per 1,000 people whereas the OECD (Organisation for Economic Co-operation and Development) average is 4.2. This leads to the interpretation that the NHS has less capacity when contrasted with other health systems. The fewer beds per head, lengths of stay and the discharge processes all lead to higher occupancy, which leaves less room to adjust to a possible surge of patients. The capacity and unplanned escalations are not unique to the NHS but a problem common across global health systems [5].

1.1.2 Process Mining

Process Mining is a technique in which data is used to discover, visualise, observe and improve business processes. [6] By extracting event logs (which are collections of cases,

each a sequence of activities paired with its timestamp) it visualises how workflows actually operate. The hospital data fits in perfectly as the patient transfer records form a sequence of activities with a patient identifier, a location, a timestamp; hence there is no need for conceptually mapping things out. Occupancy values only tell how full the hospital was at a particular instance, but an event log tells you the route an individual patient took through it. This technique is now an established method and it has been applied widely in healthcare research [7, 8].

1.1.3 Left Shifts

A patient's pathway would typically begin from hospital admission or the emergency department, a gradual escalation towards higher-intensity care, then start to descend to step-down wards and exit at discharge. In other words, the care intensity at first rises, and then it falls. A left shift is when a patient re-escalates towards a higher intensity of care after a step-down. Left shifts are unplanned, consume scarce high-intensity capacity, break the flow of patients and are associated with worse patient outcomes. How a left shift is operationalised might vary across different studies, but this project uses a prior definition from the work of Aljebreen et al. [1].

Figure 1.1 shows a simplified conceptual care pathway based on patient transfers, highlighting various left shifts.

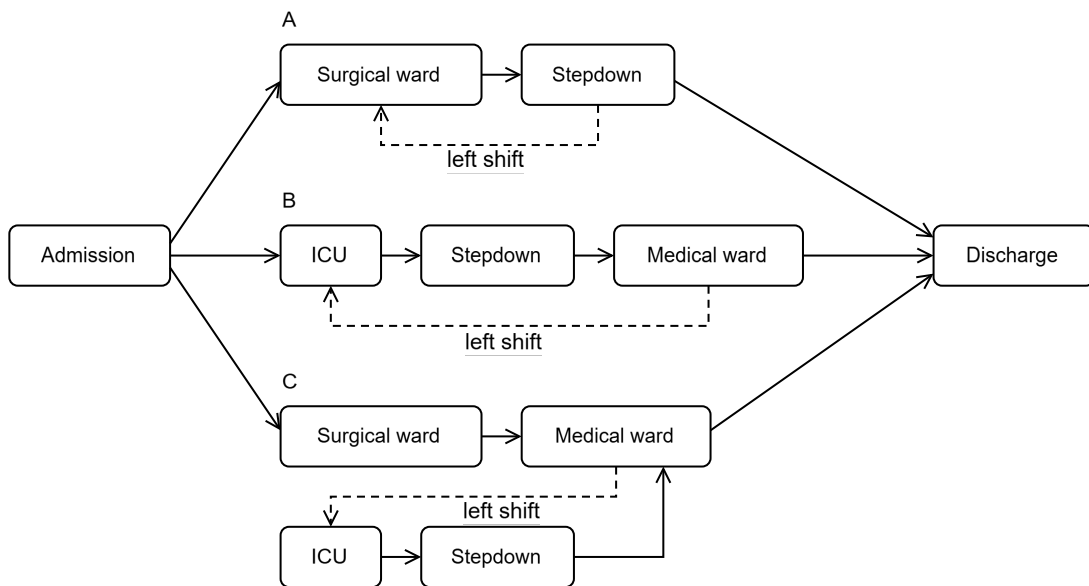


Figure 1.1: Schematic representation of a generic hospital care pathway showing left shifts. A: Return to previous unit just left. B: Return to critical care after several steps into the pathway. C: Patient exceeds earlier peak intensity.

1.1.4 Standard Techniques

There are numerous techniques adopted in healthcare research, this project utilises the framework from Aljebreen et al. [1] as the baseline and then, in addition to it, more advanced techniques are developed.

As with other process mining approaches, predictive process monitoring depends on event logs with at least a case ID, an activity label and a timestamp. Event logs on their own rarely provide enough information for reliable predictions. Optimal results require more enriched logs with additional attributes: event-level attributes that evolve across the trace, and case-level attributes that stay fixed throughout. These attributes may be in different forms, such as categorical form (e.g. admission type), numerical form (e.g. age), or textual form (e.g. clinical notes). In a healthcare setting, for example, a patient's age would stay constant during the admission; on the other hand, something like blood pressure changes is measured at multiple intervals over time.

The trace mentioned previously refers to an ordered sequence of events that connect to a single case, and it can be either finished or in progress. As PPM addresses cases that are still unfolding, the segment of a trace observed up until a certain point, before the event being predicted, is referred to as a prefix. Then these prefixes are what the models actually learn from; they are transformed into feature vectors so they can be labelled and fed into multiple classifier families, such as Machine Learning, Deep Learning, and Deep Sequence models [9]. The prediction point in this context are defined as the patient's transfer after their first ICU/SW stay (i.e., the patient's out time after their first ICU/SW visit) [1].

1.2 Objectives

1. Create a feature pipeline from the MIMIC-IV dataset.
2. Replicate the findings of Aljebreen et al. [1].
3. Develop a method that attributes predictive contribution across feature groups.
4. Build an error analysis pipeline that identifies the characteristics driving false negatives and false positives in the models.
5. Create a more generalisable framework based on care-intensity levels. The care-intensity level can be used to test whether the cohort-specific ward labels behave as their definitions imply.

1.3 Deliverables

These are the expected artefacts upon submission of the project.

1. Disco process maps of both cohorts.
2. Training code of each model in the repository (weights retained but not published).
3. The care-intensity mapping.
4. A GitHub repository with the entire codebase of the project (containing the codebase only: no patient data is included, in line with the PhysioNet data use agreement).
5. MSc Report.

1.4 Ethical, legal, and social issues

1.4.1 Ethical

In this project, the Medical Information Mart for Intensive Care (MIMIC)-IV dataset has been utilised, which is a restricted-access, de-identified electronic health record dataset. [10] The dataset has been drawn from a single US academic medical centre, Beth Israel Deaconess Medical Center in Boston, Massachusetts. Patient identifiers were removed in accordance with the Health Insurance Portability and Accountability Act (HIPAA). Date shifts were assigned to each *subject_id* so that the dates for a single patient was internally consistent. For example, if the time between two measures in the database was five hours in the raw data, then that calculated time difference in MIMIC-IV is also five hours. Conversely, distinct patients are not temporally comparable. That is, two patients admitted in the same shifted year were not necessarily admitted in the same year. This type of model indicates where clinical attention might be directed but it does not give clinical judgements. Deployed versions of any sort require clinical oversight.

1.4.2 Legal

Access to this dataset requires a PhysioNet credentialed account, with completion of the CITI (Collaborative Institutional Training Initiative) “Data or Specimens Only Research” training in human subject protections and HIPAA. It also requires signing the PhysioNet Credentialed Health Data Use Agreement. The training was completed on 9th February 2026, and the formal agreement for MIMIC-IV v3.1 was signed on 13th February 2026. Both are in Appendix D.

1.4.3 Social

MIMIC-IV introduces a single-centre scope, as all its data reflects the case mix, staffing, unit configuration, and discharge practices that are specific only to that hospital and the US system. The features that are found to predict left shifts, or how and when the model fails, are all conditioned on that setting. Models trained on historical data learn the patterns of care; in particular, *demographic features* might be the channel through which historical disparities are encoded into the model and reproduced during deployment. A further concern is that ward labels such as ICU and Surgical Ward are institution-specific and may represent different levels of care across different hospitals; therefore, care-intensity levels provide a more transferable basis for describing patient care than specific unit names.

Chapter 2

Background Research

2.1 Literature Survey

2.1.1 Unplanned escalation as a clinical and operational problem

The concept this project models is the left shift, as defined in subsection 1.1.3. The clinical literature does not use this term but it refers to the same medical event that has been widely studied which is the unplanned readmission to ICU after being stepped down. That literature is used as the baseline here and extending it to left shifts at other levels of care, which is an interpretation and not something the literature demonstrates.

A meta-analysis of 23 studies of surgical ICU patients reports a pooled incidence of 6% (95% CI 0.05–0.06) [11]. A different analysis of 229,375 admissions across 97 units in 35 US hospitals reports 6.1% [12]. The two studies are similar and are within 0.1 percentage points of each other despite having different populations and cohorts separated by more than a decade.

That convergence hides the substantial variation across both the studies. Amongst the studies included in the meta-analysis of Lu et al. the incidence ranges from 2% to 18% and differs by surgical sub-specialty, from 2% in neurological admissions to 9% in gastrointestinal surgery ($p=0.024$). The review reports that there is no significant difference in readmission rates between the surgical and other patient groups, allowing it to be used as a defensible study for general incidence estimates.

Readmission is associated with worse patient outcomes, although size depends on how the estimate is defined. In surgical cohorts, the combination of crude study-level odds ratios gives an unadjusted mortality Odds Ratio (OR) of 8.44 (95% CI 6.35–11.21, $I^2 = 89.5\%$) [11]. However, adjusting for case-mix with APACHE IV (Acute Physiology and Chronic Health Evaluation) reduces the unadjusted 5.7-fold increase to approximately four times, with a standardised mortality ratio of 3.9 [12]. Although the estimates differ from each other, both studies show a strong association, with the estimates falling between the two- to ten-fold range reported in earlier studies [11, 12].

Both readmission and post-ICU mortality vary with how the patient was discharged as well as with the patient's condition. The meta-analysis study of 18 cohorts that covers 1,191,178 patients finds that discharge outside normal working hours is associated with a risk of in-hospital mortality of 1.39 (95% CI 1.24–1.57, 16 studies) and of ICU

readmission of 1.30 (1.19–1.42, 11 studies) [13]. The reason behind this could be that the out-of-hours discharge selects patients who are no longer expected to improve from intensive care. This same study argues against this reason that if out-of-hours discharge only selected patients who are not expected to improve, then that group would show fewer readmissions, but that is not the case as they show more readmissions [13]. This interpretation still remains disputed.

Kramer et al. report that there is no significant difference in the mortality rate between night and day discharges (21.1% against 22.1%) or even between weekday and weekend discharges (22.8% against 21.4%) [12]. The reason for this difference is not resolvable from the published accounts. The explanation that the authors give for the positive findings in other studies is the differences in ICU bed availability and how the care is structured around it.

Two other findings from the same cohort support this interpretation. Discharge to a step-down unit was the independent risk factor for readmission (multivariate OR 1.17, $p < 0.001$), while a multivariable model built on demographics, clinical and physiological variables, which has acute physiology score at discharge, reaches an AUROC of only 0.67. Based on these findings, the authors conclude that patient-level factors cannot completely be the factor for readmissions [12].

2.1.2 Process Mining and predictive process monitoring in health-care

Munoz-Gama et al. [14] explain the gap between what the hospital records and what clinically happens is a characteristic of the domain rather than a unusual error. Their ninth characteristic states that healthcare records contain sensitive and low-quality data, such as missing events, imprecise and incorrect timestamps, because recording depends on manual entry by a clinician or administrative staff. As a result, the timestamps recorded might not actually correspond to the time the activity was performed. Clinicians also develop distinctive recording habits such that different patterns appear across the staff while working on the same process [14, §3.9, p. 9]. The related challenge, C6, extends this issue to data extraction and management, explaining that the uncertainty in the data should be stated explicitly and not left to interpretation [14, §4.6, p. 11].

Both of these issues are relevant to a left-shift label, as the labels are created from the recorded unit-transfer event and by virtue the label inherits whatever gap exists between the timestamps of the recorded transfer and the moment that clinical escalation happened. Prediction itself enters the framework only once, under Challenge 2, “Discover Beyond Discovery”, where predictive process monitoring is mentioned in a single sentence with a single citation without any of its characteristics explained [14,

§4.2, p. 10].

Two systematic reviews provide snapshots of the literature at different points in time. Rojas E. et al. [8] covers 74 case studies, with no lower date bound and a search completed in February 2016, while Aversano et al. [7] cover 53 journal articles published between 2014 and 2024. Prediction is not visible in Rojas E. et al. where only one out of the 74 studies use a “Prediction Algorithm” [8, Table 3, p. 230], even though the review identifies prediction as one of its four generic question types as “What will happen?” [8, §3.3, p. 229]. In Aversano et al. only 6 out of the 53 articles use “Predictive Techniques” across the three macro-categories, which is approximately 11.3% [7, Fig. 7, p. 18]. Thus, prediction was rarely implemented in the literature of 2016 and even remained scarce in the literature of 2024.

Outcome-oriented pipelines require three key decisions before a classifier is selected: how the prefixes are extracted and filtered, how traces are separated into buckets, and how sequences are encoded, where all of this takes place in the offline phase [9, §4, 17:7–17:8]. Training on every prefix is computationally expensive and can bias the model towards longer cases because they would generate more prefixes [9, §4.1, 17:9]. Across six of the benchmark datasets used, the “aggregation” encodings give an AUC which is 0.07 to 0.12 higher than the “last state” encodings. These differences were calculated from Table 8 [9, Table 8, 17:24] and not directly reported in the paper [15, §4.1, pp. 329–331].

The choice of a classifier has a comparatively small effect. XGBoost performs best on 15 of 24 datasets, but when compared with Random Forest and Logistic Regression, the performances are similar. From the above classifiers, only SVM performs significantly worse [9, §5.3, 17:23] [9, Fig. 5, 17:25]. Overall, these results suggest that decisions about how the data can be represented and divided may have a greater impact on prediction performance than the choice of a classifier itself.

Nine out of the 24 prediction tasks in the benchmarks used by Teinemaa et al.’s come from healthcare event logs [9, §5.1, 17:15–17:17]. These are convenience datasets within a domain-general study and have no clinical framing or validation. In the Sepsis-2 task, the outcome is whether a patient is eventually admitted to intensive care. It is defined as a binary case-level outcome based on the movement to a higher-acuity unit, with a positive class ratio of 0.14 and a best AUC of 0.85 [9, §5.1, 17:17] [9, Table 6, 17:18].

Clinical applications outside these benchmarks do exist but they focus on a different prediction task. Aravazhi et al. predict the next activity and its timestamp for emergency surgical patients [16, §3.1, p. 3], distinguishing this type of proactive monitoring from outcome prediction [16, §2, p. 2]. Across the studies reviewed here, none applies outcome-oriented predictive process monitoring to an ICU or acuity-transition target, and none use MIMIC as well. The studies discussed in the next subsection are those that come close to this setting.

2.1.3 Prior work predicting escalation

Five studies from 2018 to 2025 address the prediction of unplanned returns to higher-acuity care after a step-down. The outcome is described using different terms across the studies: ICU readmission [17, 18, 19], unplanned hospital readmission [20], and unplanned reoperation [21]. Although these outcomes are described differently, they are the same level of problem but at different levels in the hospital (unit-level, hospital-level, and procedure-level).

Four out of the five studies use MIMIC: Inan et al. and Chiu et al. on MIMIC-III, Aljebreen et al. on MIMIC-IV, and Rojas et al. on MIMIC-III v1.4 only as an external validation cohort only. Su et al. are the exception, using single-centre records from Tongji Hospital, Wuhan from January 2012 to December 2022. Rojas et al.’s derivation cohort comes from the University of Chicago Medical Center ranging from November 2008 to January 2016. Both MIMIC releases are from the same hospital, Beth Israel Deaconess Medical Center. Thus, the three primary studies and the validation cohort of Rojas et al.’s are from the same hospital. This limits how independently their findings can be interpreted: case mix, staffing, unit configuration and discharge practice are held roughly constant across it. Agreement between these studies therefore does not provide independent evidence that the findings can be generalised to hospitals in general.

Table 2.1: Data source, cohort, feature representation and modelling approach of the five primary studies.

Study	Data source	Cohort size	Prev.	Process features	Model families	Headline metric
Inan et al. (2018)	MIMIC-III, version not stated	Approx. 11,000; unit not stated	Not stated	None; ICU length of stay only	Naïve Bayes, C4.5, MLP	MLP AUC 0.874; accuracy 95.0 %. No external validation
Su et al. (2024)	Single centre, Wuhan, 2012–2022	2,384 patients; 83 positive	3.5 %	None; operative time only	Random forest, XGBoost, logistic regression, SVM, k -NN	Random forest AUC 0.889. No external validation
Chiu et al. (2024)	MIMIC-III, version not stated	36,232 admissions; 3,836 positive	10.59 %	None; two length-of-stay variables	BERTopic-LSTM, SVC, random forest, four boosting variants	BERTopic-LSTM AUROC 0.799. No external validation
Aljebreen et al. (2025)	MIMIC-IV, version not stated	325,119 admissions; 80,512 positive	25 %	Yes; transfer control-flow, activity counts	Logistic regression, decision tree, random forest, GBM, neural network	Neural network AUROC 0.87, no process features. No external validation
Rojas et al. (2018)	Single centre, Chicago; MIMIC-III v1.4 external	24,885 ICU-to-ward discharges	11 % (8.2 % external)	Timing and trends only; no control-flow	Gradient boosted machine, two specifications	Full GBM AUC 0.76. Externally validated on MIMIC-III, AUC 0.71

The corpus also mixes patients, discharges and admissions as if they were the same interchangeable denominators. Rojas et al.’s Table is headed “Patients (N=24,885)” while the results section reports roughly 18,000 patients. Their reported 11% therefore

is calculated as per discharge and not per patient. Inan et al. report “almost 11,000” without specifying the unit at all. Prevalence varies sevenfold across the studies (Table 2.1), which is important when interpreting the performance values below, since threshold-dependent metrics like precision are directly affected by prevalence.

Aljebreen et al. are the only study to use control-flow information: 77 transfer activities are collapsed into five, producing 942,368 events and 2,423 trace variants, with aggregation encoding with static case attributes. Rojas et al. use ICU length of stay and trends in vital-sign and laboratory results over the 24 hours before discharge but do not use sequences. For all other predictors, they use the measured value closest to ICU discharge. Inan et al. use 18 variables based on averaged laboratory results from the last 24 hours, together with demographics values, with ICU length of stay as the only duration measure. Chiu et al. use 22 structural predictors with ICU and hospital length of stay. They also use unstructured text through BERTopic topics derived from discharge summaries but do not use sequences. Su et al. use preoperative and intraoperative variables with operative time as the duration measure. Overall, the order of events is not used in four of five studies.

Classical ML dominates these studies. Only Chiu et al. include a sequence model, while Rojas et al. use gradient-boosted machines alone, comparing their results against SWIFT (Stability and Workload Index for Transfer) and MEWS (Modified Early Warning Score). These are published rule-based scores rather than classifiers they fitted to the data. The best reported AUC values range from 0.76 to 0.889. However, these values come from different outcomes and units of analysis, so ranking the studies by AUC would not be meaningful. For example, Su et al.’s 0.889 is reported for an outcome with a prevalence of 3.5%, whereas Aljebreen et al.’s 0.87 is reported for an outcome with a prevalence of 25%.

Only one of the five studies includes external validation: Rojas et al. validate their model on MIMIC-III v1.4, where the AUC falls from 0.76 internally to 0.71 externally. The model transferred to the external cohort was the reduced physiological GBM and not the full model because nursing, medication and ICU-intervention variables were not available in MIMIC-III. Chiu et al. explicitly identify the lack of external validation as a limitation. Su et al.’s describes their held-out set as “an independent testing cohort”, but it is a 7:3 split from the same single-centre database. Aljebreen et al. use a 67/33 split and note that chronological splitting is not feasible because MIMIC-IV is temporally anonymised. The studies also use different definition of the outcome itself, where the difference is explained in Section 2.1.6.

2.1.4 What the reviews report about the field

The literature on predicting unplanned escalations has been synthesised in two reviews with differing scopes and methods. Ruppert et al. [22] review ICU readmission prediction across different model classes, covering 33 studies published between 2010 and 2021, and organising their findings around five methodological recommendations. Koumantakis et al.’s [23] in contrast, focus only on DL models and carry out a systematic review and meta-analysis of 24 studies covering a total of 49 models. The two reviews therefore use different approaches. Ruppert et al. provide recommendations about how studies should be designed, while Koumantakis et al. report pooled estimated and heterogeneity statistics (Appendix A). This difference is partly a consequence of the studies themselves, as Ruppert et al. considered the included studies too heterogeneous for meta-analysis.

The two reviews reach similar conclusions on the issues they both examine. Ruppert et al. find that the higher-performing models tend to use more homogeneous cohorts, often post-surgical, and longitudinal predictors rather than relying on a particular model class. Similarly, Koumantakis et al.’s subgroup analysis reports a pooled AUROC of 0.92 (95% CI 0.88–0.94) across the two disease-specific subpopulation studies, compared with 0.79 (0.73–0.85) across all eleven studies. Heterogeneity also falls substantially, from 99.96% to 17.1%. Both reviews point to a similar dependence on a small number of US data sources (section 2.1.3). Ruppert et al. report that 25 of their 33 studies used data from a single centre.

External validation is rare in both reviews. Six of Ruppert et al.’s 33 studies were externally validated, compared with the two of 24 studies included by Koumantakis et al.’s. Both reviews also report that discrimination decreased when models were tested on external data. Despite being separated by differences in scope and method, as well as four years of additional literature, the two reviews describe the field in the same way.

Ruppert et al. identify outcome definition as one of their five methodological recommendations, under the heading *Be Specific When Defining Outcomes*.

Koumantakis et al. report similar variation across their studies: 16 of the 24 studies define readmission as occurring within 30 days of ICU discharge, five define it as readmission during the same hospital stay regardless of time window, and one uses a 48-hour window. A further six examine mortality alongside readmission. These differences are also included in the analysis as a subgroup variable. Among the seven 30-day studies eligible for pooling, including mortality changes the pooled AUROC from 0.70 (0.63–0.77) across five studies to 0.79 (95% CI 0.78–0.80) across two, while I^2 falls from 99.9% to 0%. Eight studies (33%) use a composite outcome that combines readmission with death rather than reporting mortality separately, and these are recorded as an applicability concern.

2.1.5 Label quality and construct validity in clinical ML

A clinical prediction task involves two separate things: the event a study aims to predict and the label in the data used to represent that event. Whether the label actually represents the intended clinical event is a question of construct validity. Whether the labelling rule has been applied correctly is a question of label quality. These are separate issues, and either one can fail without the other.

Ghassemi et al. [24] note that diseases are often poorly labelled in the electronic health records, which are designed to support care and billing rather than analysis [24, p. 191]. Their example of a pneumonia diagnostic code illustrates a validity problem rather than an accuracy problem: a code may indicate that a patient was screened for pneumonia rather than that the patient actually had pneumonia [24, p. 193]. They also warn against using clinician behaviour as a label source because clinicians may be wrong [24, p. 193]. In this case, the labelling rule may be applied correctly, but the resulting label can still fail to represent the clinical event, making this a validity issue rather than an accuracy issue. Ghassemi et al. also describe a medical definition as a working model of the current understandings that can change as that understanding develops. They illustrate this with acute kidney injury, for which both RIFLE (Risk, Injury, Failure, Loss of function, End-stage renal disease) and KDIGO (Kidney Disease: Improving Global Outcomes) definitions are used [24, p. 193].

Hagmann et al. [25] describe a failure mode that occurs when three conditions are met: the label is generated by applying a labelling function to a set of fundamental measurements, those same measurements are included in the model input, and the labelling function can be approximated by a generalised additive model [25, p. 11]. In this setting, the model learns the labelling function rather than other predictive patterns in the data. The measurements used to define the label develop non-zero fitted shapes, while the remaining features converge towards a constant zero contribution [25, p. 11]. As a result, the model can reconstruct the label almost exactly and perform well on test data generated in the same way, but its performance breaks down when the defining measurements are missing or incomplete because it has not learned any alternative predictive patterns.

Hagmann et al. distinguish this failure label leakage, where information unavailable at prediction time is introduced into the model. In their case, each use of the measurements is legitimate on its own; the problem comes from using the same measurements both to define the label and as model inputs [25, p. 8]. This differs from Ghassemi et al., who classify comparable cases as leakage [24, p. 193]. The resulting problem is structural rather than something that can be fixed through more careful labelling. In fact, a more faithful implementation of the definition makes the reconstruction even more exact. The issue therefore lies in the relationship between the

label and feature set, rather than in how accurately the label itself was constructed.

2.1.6 Research gap

The five studies reviewed in Section 2.1.3 do not define the outcome in the same way. In Rojas et al. [19], there is no time window; a positive is any readmission occurring during the same hospital stay. Su et al. [21], by contrast, use a 30-day window, but determine whether a case is positive based on anaesthesia type rather than the destination unit. This shows that the disagreement is not simply about how long the outcome window should be, but also about what counts as the event itself. Across the five studies, the definitions differ in several aspects, including the time window, whether the destination unit is part of the definition, whether death counts as a positive outcome, and whether planned returns are excluded or treated as negatives. Since these definitions capture different events, their reported prevalence and performance measures are not directly comparable.

In several of these studies, the rule used to generate the label does not fully match the event described by the paper’s outcome name. Chiu et al.’s [18], for example, define their outcome as ICU readmission, with a reported readmission rate of 10.59%. However, their selection rule includes patients who were either readmitted or died within 30 days of ICU discharge [18, §2.2, p. 5]. In their cohort table [18, Table 1, p. 10], 774 patients in the positive group (20.18%) have a discharge location recorded as Dead/Expired. Patients who died during the hospitalisation were excluded from the cohort [18, §2.2, p. 4], whereas deaths occurring after discharge were counted as positive outcomes. The paper does not clearly define where the boundary between these two groups lies.

Aljebreen et al.’s conceptual definition is an emergency return for the same medical condition [20, §1, p. 422], whereas the implemented rule uses any emergency admission within 30 days, regardless of the cause. The authors acknowledge this as a limitation, noting that MIMIC-IV does not provide enough information to apply the full definition. In all of those studies, the label-generating rules are visible, but that visibility does not show that the rule actually captures the intended clinical event. Therefore, disclosure and validation are different things.

Across all five studies, there is no empirical check reported on whether the label actually measures the intended clinical event. There is no chart review, outcome correlation, or clinician adjudication. Rojas et al. perform a post hoc sensitivity analysis in which the excluded comfort-care group is reclassified and the model is re-fitted. The resulting AUC is 0.75 compared with the 0.76 for the primary analysis [19, Model Development, p. 848; Results, p. 850]. This tests whether model discrimination remains stable when the label definition changes, which is different from testing whether the label captures the intended event. Su et al.’s describe three-author manual data collection with

Table 2.2: Stated outcome name against the operational positive-label rule, with reported label validation.

Study	Outcome name	Positive-label rule	Label validated
Inan et al. (2018)	ICU readmission	No operational rule stated: no time window, no unit or transfer rule, no exclusions; readmission is tied only to the same hospitalisation	None stated
Su et al. (2024)	Unplanned reoperation	Reoperation under general, spinal or epidural anaesthesia within 30 days of the index surgery for medical necessity; excludes surgery prompted by pathological findings	None stated
Chiu et al. (2024)	ICU readmission	Readmitted <i>or</i> deceased within 30 days of ICU discharge; the unit of return is not specified	None stated
Aljebreen et al. (2025)	Unplanned hospital readmission	Next admission within 30 days of discharge and of emergency type; the same-condition requirement in the paper’s own definition is not implemented	None stated
Rojas et al. (2018)	ICU readmission	ICU-to-ward-to-ICU transfer at any point in the same hospitalisation (no window); ward cardiac arrest also positive; post-procedure readmissions retained as negatives	Partial

consensus [21, §2.2, p. 2] covering “all clinical variables”, but do not state that this process was also used to establish the unplanned reoperation outcome. Table 2.2 shows the stated outcome name of each study against the rule that generates the positive label and if any validation checks were done.

Ruppert et al.’s [22] make the label-validity issue explicit in a section titled “Be Specific When Defining Outcomes”[22, p. 5]. They note that readmission can result from factors unrelated to the patient’s condition at discharge, including nosocomial infection, administrative causes, or deterioration after leaving the ICU [22, p. 5]. They also cite a multicentre cohort in which fewer than half of readmissions during the same hospitalisation were related to the initial complaint [22, p. 5]. Based on this, Ruppert et al. argue that “Operational and administrative factors decouple ICU readmission decisions from the underlying patient physiology”, which forms one of their five recommendations [22, p. 5].

Koumantakis et al.’s [23] review of 24 studies and 49 models approaches the same variation differently. They count the different outcome definitions (16/24 at 30 days, 5/24 using the same hospital stay, 1 using 48 hours, and 6 including mortality), incorporate them into subgroup analyses, and report them as a source of heterogeneity ($I^2=99.96\%$) [23, p. 3; pp. 7–8]. Studies using mortality as part of a composite outcome are also recorded as an applicability concern ($n=8$, 33%) [23, p. 7]. However, the review does not examine whether recorded readmission is a valid proxy for deterioration. Thus, the validity problem is recognised in the review literature, but none of the five primary

studies test it, while the most recent synthesis treats the variation as a source of statistical heterogeneity rather than as a threat to validity.

Anyone predicting this outcome has to choose a definition, yet the dataset provides no clear basis for choosing one. What remains unclear is how much the choice of definition alone changes the results when applied to the same cohort.

2.2 Methods and Techniques

This section describes the main methods available for predicting this outcome, with section 2.3 specifying those used in this study. The main decision is whether to represent a stay as a feature vector at the prediction point or as an ordered sequence of events with prediction made from a prefix. The conventional tabular approach offers mature tooling and results that can be compared directly with existing clinical prediction studies. Its main limitation, however, is that two patients with the same values at the prediction point appear identical to the model, even if their prior histories are very different.

The process-aware approach makes the patient’s route visible, but this requires reconstructing an event log from the transfer tables. As this reconstruction inherits the recording gap mentioned in subsection 2.1.2, the trace is more an inference than a record.

2.2.1 Event log construction and activity abstraction

Constructing an event log requires three components for each event: a case identifier to group events into traces, an activity label to describe what occurred, and a timestamp to order events within a trace (subsection 1.1.2). Hospital transfer records fit this structure reasonably well, as each row represents a patient’s movement into a care location, with the location providing the activity label and the recorded time providing the timestamp. However, neither the case identifier nor the activity label is explicitly provided by the data. A case can be defined either as a single admission or as an individual patient, producing a separate trace for each hospital stay or one trace covering the patient’s entire history. This choice also determines whether a readmission appears as an event within an existing trace or as a separate case.

The activity label is therefore constructed rather than simply taken from the record, and how it is constructed is itself a modelling decision. Locations can be kept using the raw transfer vocabulary, preserving every distinct unit recorded by the hospital, or grouped into a smaller number of care categories. Published work uses approaches across this range (subsection 2.1.2). A finer-grained vocabulary keeps clinically different locations separate, but creates more distinct traces and make control-flow encodings based on activity sequences sparser (subsection 2.2.2). A coarser vocabulary produces denser encodings and fewer trace variants, but can also remove distinctions that may be

clinically meaningful. Neither choice is correct; the appropriate level of granularity depends on which distinctions the prediction task needs to capture.

Unit names are not the only way to represent care location. A care-intensity scale instead orders locations by the level of care they provide, making the label ordinal and potentially more comparable across institutions with different unit names. Such a scale can either be taken from a published standard or derived from the data itself. A published standard provides external authority, but it describes the level of care a patient requires rather than the location they occupy. Applying a UK standard to US hospital data also introduces a jurisdiction mismatch. A data-derived ordering fits the vocabulary of the dataset more closely, but has no external basis and may not transfer beyond the data from which it was derived. With either type of vocabulary, and at either level of abstraction, the recorded timestamps may still not correspond to when the activity actually occurred (subsection 2.1.2).

2.2.2 Pipeline design decisions

A process-aware pipeline takes three decisions before any classifier is selected. They are how the prefixes are extracted from traces and which are retained, whether traces are partitioned into buckets, and how a prefix is encoded as model input. Benchmark evidence indicates that these decisions influence outcome prediction more than that of a classifier choice (subsection 2.1.2). The following sets out which options within each are available for reconstructing an event log from transfer records.

A prefix is the portion of a trace that is available at the moment a prediction is made (subsection 1.1.4). Training on every prefix of every trace is expensive and heavily biases the training set towards the longer cases, as a trace of many events creates proportionally as many prefixes. By filtering the prefixes at a fixed event count, at fixed gaps along the trace, or by execution time, training cost reduces as well as the length bias, but at the cost of coverage of the prefix space. Where the prefix ends with respect to the prediction point is a further decision. A prefix that includes the event at the boundary can include information about the decision being predicted (subsection 2.1.5).

Rather than training one classifier on all prefixes, bucketing first groups the training prefixes and then trains a separate classifier for each group. The different approaches vary in how they define those groups: prefix-length bucketing groups prefixes by the number of events observed, clustering groups them by similarity in the encoded feature space, and state-point or decision-point bucketing groups them by their position in the process. Depending on the approach, the bucket may need to be determined at prediction time or may follow directly from the prediction point. Therefore, bucketing increases the number of models, with each model trained on only part of the data, and adds a runtime assignment step on which the prediction then depends.

A prefix is a variable-length sequence, but a classifier requires a fixed-length input. Encoding provides the mapping between the two. Last-state encoding uses the attributes of the most recent event and discards the earlier events. Aggregation encoding summarises the entire prefix, retaining its content but losing the order of events. Index-based encoding preserves order by assigning separate features to each position in the sequence. Learned representations instead learn this mapping from the data during training. Sequence models use integer-encoded, padded sequences, which preserve the event order but require a fixed maximum length, while also the padding carries no information.

Two control-flow encodings can be used for activity sequences based on care location: n-gram counts, which captures order within the chosen time window but not beyond it, and integer sequences passed to a sequence model, which preserve the order throughout. The level of abstraction chosen in subsection 2.2.1 affects these options, since a vocabulary with many distinct activities leaves n-gram counts sparsely populated.

2.2.3 Choice of classifier

Several classifier types can take an encoded prefix and a binary outcome. Benchmark evidence suggests that the choice between them has less effect on performance than the pipeline decisions above (subsection 2.2.2). They differ in what they require rather than what they achieve. Sequence and neural models have more parameters and generally require more positive examples, which is important when the outcome has a low positive rate. They are sensitive to initialisation, so a single training run does not necessarily represent the behaviour of an architecture.

They also differ in how easily their predictions can be interpreted, ranging from directly readable linear coefficients, through post-hoc attributions for tree ensembles, to learned representations that require separate explanation methods. When a prediction is intended to inform a step-down decision (subsection 1.1.1), inference latency becomes part of the model selection rather than simply an implementation detail. Every classifier type also has hyperparameters, so comparing different types is partly a comparison of how each has been configured. Choosing these settings through a search also uses training data that can no longer be treated as unused data.

2.2.4 Analysis of fitted models

A performance metric shows how well a model predicted the cases, but not which features drive that performance or which cases it gets wrong. Feature importance can be examined by changing the feature set, for example, by removing groups of features, adding them, or using them alone, while keeping the classifier fixed. Any change in performance can then be attributed to the features rather than to a change in model

capacity. Test cases can also be labelled by their confusion-matrix categories and their feature distributions compared across categories, showing where the model performs well and where it fails. Both methods depend on the classifier selected, and correlated feature groups cannot be separated cleanly into independent contributions.

2.2.5 Evaluation design

Where the outcome occurs in only a small proportion of cases, the behaviour of performance metrics changes with the positive rate, which affects what comparisons they can support. Threshold-dependent metrics depend on both the positive rate and the threshold used to calculate them. Since the threshold is not often stated in the studies, the choice of metric needs to be made before comparing results rather than after. AUROC does not change with the positive rate, but this does not make AUROC values directly comparable across studies with different outcome definitions, case mixes, or unit of analysis. AUPRC has a no-skill baseline equal to the positive rate, so its value cannot be interpreted without knowing the threshold rate, and values from cohorts with different positive rates are not directly comparable.

When training involves randomness, a single model fit is one draw rather than a fixed property of the method. Repeating the training across different seeds makes it possible to distinguish differences between models from the variations within each model. The way the data are divided is also an evaluation choice rather than simply a preprocessing step. Splitting by patient makes sure that records from the same patient do not appear on both sides of the split, while also using the same test partition for every model makes sure that the performance differences are not caused by different cases being included in each test set.

2.2.6 Defining and validating the outcome

The outcome, much like the activity label, is computed from transfer records by a rule and not directly observed from the data. That rule is a design decision equivalent to those discussed previously. Where subsection 2.1.5 asked what happens when a label is constructed badly, this subsection sets out which construction and validation options are available for a label that are derived from transfer records. This rule defines the prediction target preceding and fundamentally shaping the subsequent choices regarding encoding, classifiers, and evaluation.

Unit-name rules define the outcome as a return to a specific named unit types. Because the rule and the record use the same vocabulary, the implementation can be checked directly against the data. However, units that have different names but provide similar levels of care may not be treated as equivalent, and the rule may not transfer to a hospital that uses different unit names. Time-windowed rules instead define the outcome

as a return within a fixed period of time, or within the same admission with no specific time period given. The length of the window decides which returns to the hospital are treated as connected to the original discharge: a short window might remove late deterioration, while a long window might include events that are completely unrelated to the task.

Acuity-level rules define the outcome as a re-ascent on a care-intensity scale, which makes this definition independent of local unit names. However, mapping a location to a level requires a judgement rather than a simply reading the values from the records, so the label inherits the limitations of that mapping, as discussed in subsection 2.2.1.

Route-conditioned rules instead define the outcome as what occurred between departure and return, such as whether the patient left the hospital, died, or passed through intermediate locations. This separates readmission from escalation within the same stay, but it also creates more rule variants, and with the conditions increasing, the definition can drift away from the original outcome.

These four dimensions are independent design choices rather than mutually exclusive options. A single rule will fix more than one of them at the same time, which makes a set of variants a coherent set of definitions rather than an arbitrary list.

A definition can be checked, but the available approaches are different in what they can establish and in the resource they require. Chart review and clinician adjudication can test whether the event the rule labels corresponds to the clinical event it is meant to capture, but they require access to records and clinician time. Correlation with downstream outcomes is less costly and can test whether the label behaves as expected. However, a label that captures something correlated with the event rather than the event itself could produce the same pattern. Internal checks comparing the stated definition with the implemented rule are cheaper still, but they establish only whether the rule has been implemented correctly or not. A rule can be implemented correctly but it still can represent the wrong clinical event.

2.3 Choice of methods

This project represents the patient stay as an ordered trace of events and predicts from a prefix and not a feature vector at the prediction point. Four out of the five prior studies on this outcome work in tabular form (subsection 2.1.3). This means there is no direct comparisons with that literature and to treat a trace as an inference from the transfer records rather than what actually occurred.

Locations are grouped into a small set of clinical categories from the raw transfers table and are labelled by care setting rather than by local unit names. This means that the representation captures the movement between different setting and not by the

movement between the individual named units; two units of the same type are treated as the same event for this outcome. The raw transfers table vocabulary was rejected because it multiplies trace variants and increased the control-flow encodings (subsection 2.2.1) without adding anything that the outcome depends on. The grouping is itself a judgement about which locations count as the same setting rather than an interpretation of the record. It therefore also removes the distinctions that would have been retained by the finer-grained vocabulary.

Prefixes are bucketed by decision point as explained in subsection 2.2.2, with a separate event log, feature matrix, partition split and classifier for each of the two cohorts. Now, cohort membership is determined by the prediction point rather than being assigned at runtime. The two buckets overlap, as the prediction points are defined independently and a single admission might reach both. Prefix-length bucketing and clustering over prefixes were not used. Aljebreen et al. [20] report that clustering with two clusters did not affect their results, so adding it would create a parameter without evidence of benefit.

Multiple classifier models are compared, which are a selection of both tabular and the sequence-consuming model groups as discussed in subsection 2.2.3. According to the benchmark evidence, classifier choice impacts the results less when compared to the pipeline decisions. None of the five studies on this target uses a model that consumes the sequence directly (subsection 2.1.3), so this comparison tests whether the broader benchmark findings also hold for this study rather than just assuming that they do. The trade-off with this is that a larger set of models are to be trained and evaluated, with no single model used as a concrete result.

AUROC is the primary metric used in this study as explained in subsection 2.2.5. It does not change with the positive rate, making it suitable for comparing label variants within the same cohort. AUPRC is reported alongside it and interpreted against the no-skill baseline given by each positive rate of each variant. Threshold-dependent metrics are calculated at a stated threshold rather than an implicit one. The data is split by patient so that no patient appears on both sides of the split. The per-patient date shift (section 1.4) means that admissions of different patients cannot be ordered against one another from the timestamps. As a result, chronological splitting is not available from the event data.

The prediction outcome is defined over the same care-unit grouping of ICU and SW activity categories. The primary definition fixes two of the dimensions of subsection 2.2.6, which are the unit-name parameters to those groupings and the time parameter to same-admission with no upper bound.

This project reports the results under a set of definitional variants rather than under a single rule. The variants differ along two axes, one being that the destinations count as

a return and the other being what counts as a movement between them. The cohort, feature set and pipeline are held fixed throughout the whole execution. This method does not test sensitivity to a return time window. It also does not establish that any of the definitions capture the clinical event, which would require the validation methods of subsection 2.2.6 rather than a robustness check alone.

The acuity-level labels described in subsection 2.2.1 is not used as the primary outcome, being a separate method of analysis introduced in Chapter 3. Chart review and clinician evaluation were not undertaken because both require access to records beyond the dataset and as well as clinician time was not available for this project.

The main design decisions were all focused on the prediction outcome definition, prefix boundary and the evaluation design rather than hyperparameter tuning within any one of the models. The following chapter implements these design decisions and reports the resulting comparisons.

Chapter 3

Datasets and Experimental Design

3.1 Datasets and data sources

3.1.1 Data source, access and extraction

MIMIC-IV v3.1 is the data source for this project, obtained from PhysioNet under credentialed access. The training requirement, credentialing process and the data use agreement are all stated in subsection 1.4.2.

Aljebreen et al. [1] do not state its used release version of MIMIC, as this omission according to subsection 2.1.6 is more of a characteristic of the field. The cohort definition of this study cannot be checked against the source paper’s one as the release it was made on is not stated, and subsection 3.1.4 explains the impact of it on this study.

The database extraction was verified by row count before any further processing was done. This v3.1 release of MIMIC-IV contained a total of 546,028 hospital admissions and 364,627 patients. Out of the total patients, 223,452 distinct patients have at least one hospital admission. The remainder of the patients appear in the module but do not have a corresponding hospital stay, meaning they have no transfer events and do not enter the final cohorts. From the MIMIC database, a total of seven tables were used and are all explained in Appendix C.1. *PostgreSQL* was the database tool used to load the tables and query from.

The table most frequently used out of the seven is the *transfers* table as it is the main foundation from where most of the subsequent parts of this project is derived from and are listed as follows:

- Event log construction (subsection 3.1.2)
- Prediction points (subsection 3.1.4)
- The dependent variables (subsection 3.1.5)
- All but one of the process features (subsection 3.1.6)

The only exception here is that the admission location is derived from the *admissions* table. The emergency department flag is set by reading the constructed event log for the ED activity, which is the indirect usage of the *transfers* table.

3.1.2 Event log construction

Out of the three components an event log construction requires (subsection 2.2.1), only the timestamp is directly read from the *transfers* table. The remaining two are constructed from the table and each construction has its own design decision.

The case ID is *hadm_id*, so one case is one hospital admission and one trace is the route the patient takes through a single hospital stay, as opposed to the case-per-patient alternative described in subsection 2.2.1. A subsequent hospital readmission registers as a new case and not as an event within an existing trace, which is consistent with the dependent variables defined within a single admission (subsection 3.1.5).

Timestamps are derived directly from the *transfers.intime* which is the recorded time of arrival at a care location, while the *admissions.admittime* provides a synthetic *Hospital Admission* anchor for every admission. However this anchor is not always the first event, as in 373,434 traces the emergency department arrival is recorded earlier, which means a total of 145,567 of the 546,028 traces record the *Hospital Admission* time first. Activity labels are formed from the combination of *careunit* and *eventtype*, which is then mapped to the eleven categories of subsection 3.1.3.

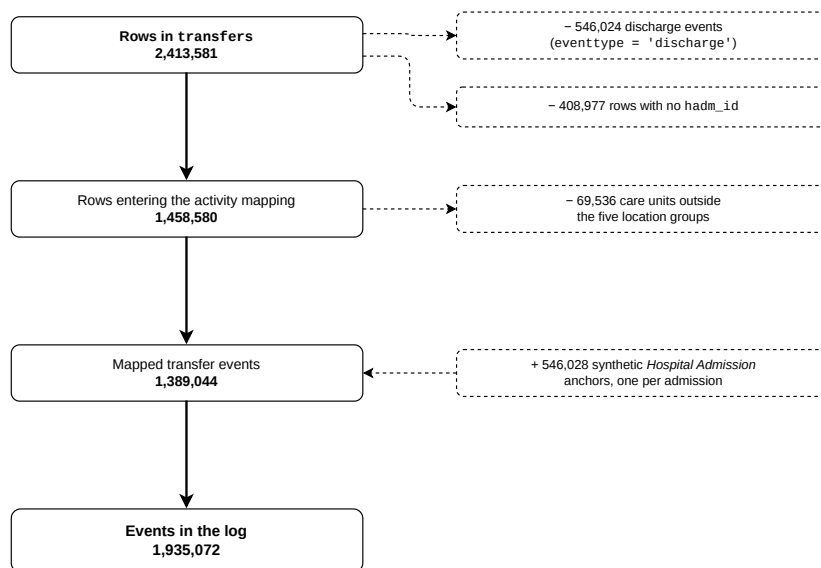


Figure 3.1: Event-log construction, reconciled row by row against the *transfers* table. The resulting log holds 1,935,072 events over 546,028 cases and eleven activities, forming 10,977 trace variants with trace lengths from 1 to 34.

Discharge events were excluded as the prediction is made before the discharge, so every trace ends at the patient’s last recorded arrival and not their exit from the hospital. Care units that were not in the five location groups were excluded, which removed nine out of the 46 distinct care units in the *transfers* table (subsection 3.1.3). Admissions and transfers records were kept as separate activities so that an arrival records whether

the patient entered the hospital there or had moved internally, which the *careunit* value does not carry that information alone.

A total of 330 rows are genuine unknowns. The *eventtype* filter and care-unit mapping remove the same rows together. The event log has 4 admissions with no *transfers* row at all, giving it a trace length of one.

The total case count equals the admission count of subsection 3.1.1. The event log in this project was constructed over the entire admissions in this MIMIC-IV release and was a divergence from the source paper’s method of creating two cohort-restricted logs. This justifies the difference in the variant counts, as here it is 10,977 compared against the 338 and 356 of the source paper. This event log hence admits every pathway recorded for the admissions, the cohort restriction is applied at cohort construction instead (subsection 3.1.4).

The per-admission event rates are similar between this study and the source paper, at 4.277 for the source against the 4.223 here for ICU and 3.156 against 3.342 in the SW cohort respectively. This is a difference of 1.3% and 5.9% even though the cohorts differ in size by 56% (ICU) and 9% (SW) against the source paper sizes. What differentiates the cohorts is which admission enters then and not how the events are extracted. On an anchor-free basis the trace lengths run from one to 33.

However the recorded timestamps do not need to correspond to when the activity occurred (subsection 2.1.2), so every event records the movement when it was entered and not when the patient moved.

3.1.3 Activity abstraction

Section 2.3 settles in favour of the aggregating by care setting method, since two units of the same type are the same event for this outcome.

The *transfers* table in the database contains 46 distinct *careunit* values, and these 46 are then aggregated into five location groups listed in full in Appendix B.1. Each group has two activities, one is to distinguish arrival by *Hospital Admission* from arrival by internal transfer (subsection 3.1.2), and the synthetic hospital admission anchor.

Figure 3.2 shows the resulting process map. Appendix C.2 shows the same model with the path filter at 100% path visibility.

Two independent placements in the mapping are judgements and not an inference of the records. The *post-anaesthesia* care unit is grouped with the *Stepdown/Intermediate* and the *Discharge lounge* with the *Medical Ward*. Out of the two, neither follows from the unit name itself, and another study may place these units elsewhere.

Care units outside the mapping

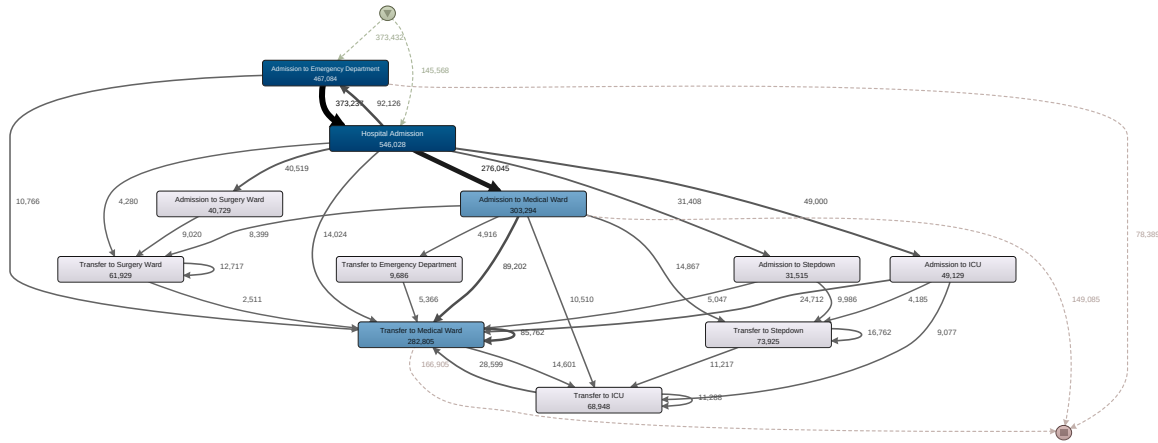


Figure 3.2: Directly-follows model of the event log generated in Disco. The path filter is set at 30% to show frequent paths.

The five distinct location groups that are made enumerate 37 out of the 46 distinct care units. The remaining nine care units are found in the database but have no trace length or clinical activity logged, and are dropped from the logs as a null activity. Their row counts are given in Appendix B.2.

Out of the nine dropped care units, six are obstetric or neonatal and follow a care trajectory that the five location groups were not built to describe, which is the same justification applied to the care-intensity mapping of subsection 3.1.8. The Medical/Surgical (Gynaecology) is a general ward that belongs in the Medical ward group but its exclusion is based on how the mapping worked and not an intentional removal. Across the seven non-unknown unmapped units that were excluded affect only a small proportion of admissions (1.44% in ICU and 0.72% in SW), and the actual number of patient sequences affected is even smaller.

The vocabulary is unordered

The eleven activity categories are clinical groups created with care intensity in mind, but these grouping are not based on a validated scale. Section 2.3 selects the care-unit named labels as the primary study which aligns with the source paper and the care-intensity level mapping as a secondary study that validates the prior. No rule in this chapter compares the two categories as more or less favourable, and the comparison is made separately against the *Intensive Care Society* mapping of subsection 3.1.8.

3.1.4 Cohorts and prediction points

Two separate cohorts are constructed, one for each prediction point explained in subsection 1.1.4, and each cohort follows the decision-point bucketing chosen in section 2.3. This makes the cases being assigned to its corresponding cohort based on

the selected prediction point and not assigned during runtime.

The ICU and SW cohorts, each are the intersection of its corresponding feature matrix and sequence file, so that every model is trained and evaluated on the same set of admissions. If the cohorts were given their individual set of admissions, this could have made performance differences as the evaluated patient populations would be different.

Age filter was not applied in the cohorts as MIMIC-IV v3.1 is a majority adult dataset, and this is validated by a direct count confirming that neither cohort has an admission for a patient under the age of 18.

When the cohorts are compared with the source paper, they differ in size and some parts of the difference is not attributable. Under the same planned-surgery exclusion in the SW cohort, this project counts 67,066 admissions against the 61,312 size the source reports (a 9% increase). The ICU cohort has no pre-defined exclusions like the SW and it comes at 85,248 admissions to the reported 54,782, which is a 56% increase in size and the excess cannot be attributed. Section 5.4 reports what was tested and what remains unexplainable. The process behaviour of each cohort is shown in Appendix C.3.

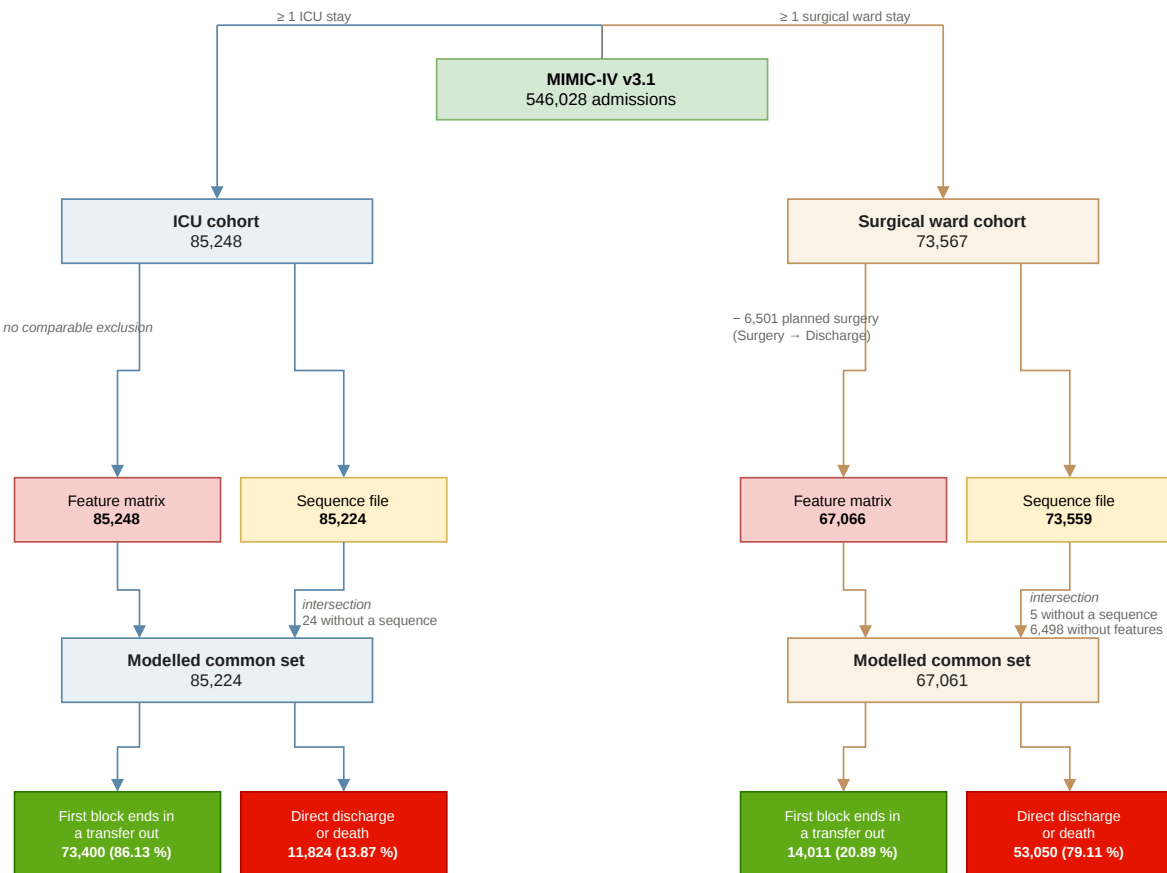


Figure 3.3: Cohort construction with each cohort as the intersection of the feature matrix and sequence file.

Admissions incapable of a positive outcome

An admission whose first stay in that unit type ends in a direct discharge or death from the hospital has no transfer out, and there is no possibility of a subsequent return. These types of admission can never become positive in any label variant. When validating this point, a direct check had confirmed that there was no positive count for such types of admissions in every case. These types of admissions were named as structurally-negative admissions.

In the SW cohort most of the negative labels used to fit the models are negative because of how the cohort is defined and not how the labels are constructed. When compared with the ICU cohorts, this was affected far less, roughly around one in seven admissions, and these types of admissions were not removed which was in accordance with how the source paper gave its cohort definition.

In the ICU cohort, a substantial amount of admission were deaths which made it a competing risk rather than any missing information. A patient who dies in the first stay cannot return to a higher level of care, so these types of admissions were recorded as negative, which was a modelling decision. The same pattern occurs in the SW cohort where there is a large group of admissions where the patient is discharged from the ward alive, and hence applying the same reasoning as we did for the ICU is not applicable here.

3.1.5 The dependent variable

The prediction outcome is calculated by applying a pre-defined rule to the transfers data and not inferred from the data directly (subsection 2.2.6). The defined label rule was that an admission is positive if the patient returns to the ICU/SW unit after its initial stay within the same hospital and no return time window. Out of the four dimensions described in subsection 2.2.6, the destination (given by unit name grouping) and the time parameter by same-admission with no window, are fixed. The label is computed directly from the *transfers* table and not from the aggregated event log because of one asymmetry. The event log describes whether the patient is a new admission or is a transfer from another unit, but here the label will treat this as the same event. This makes the label less detailed when compared to the event log which is used in this prediction.

The primary label used here is the *lat* label, as this label variant implements the prediction point as how it is described in the source paper, which is a transfer to another ward applied to both the cohorts. The label name is compressed *lateral-transfer correction* and does not indicate that lateral movements are included. *lat* uses the broader rule and remove the returns in which the patient never left the tier, which required at least one recorded stay outside it. This condition makes sure that it is

deliberately outside the tier so that the Surgery $\rightarrow$ ICU $\rightarrow$ Surgery sequence is counted as positive. Appendix C.4 shows the ICU cohort split by the *lat* label.

Table 3.1: The five label variants, with each of their positive rates on the feature matrix and test-set.

Variant	Rule	Full cohort (ICU 85,248 / SW 67,066)	Test set
old	Positive if the patient re-enters any unit of the target tier after the first stay row in that tier ends. Same-tier movement counts.	ICU 14.95 %, SW 14.26 %	ICU 15.16 %, SW 13.88 %
lat (primary)	old , less returns where the patient never left the tier in between. Requires at least one recorded stay outside the tier.	ICU 11.68 %, SW 9.52 %	ICU 11.88 %, SW 9.41 %
corr	SW only. Positive if the return is routed through none of ICU, the seven intermediate units, or PACU.	SW 8.02 %	SW 7.72 %
corrpacu	SW only. corr , plus returns routed through PACU (970 admissions).	SW 9.47 %	SW 9.19 %
collapse	SW only. corr , computed after merging consecutive rows in the same care unit into one stay, so that bed and room moves cease to count as a departure and return.	SW 2.27 %	SW 2.29 %

Aljebreen et al. [1] reports a 16.8% ICU left shift rate, and when compared with the *lat* label is 11.68%, while the *old* label has a left shift rate of 14.95% which is closer to the source paper’s value. For both *lat* and *old* labels, they were across the entire cohort which replicates the method the source paper approached. However this does not validate that to select *old* over *lat*. In subsection 3.1.4 it is established that the ICU cohort is 56% larger than the source paper’s cohort size, and that substantial difference could also not be attributed. This shows that rate agreement is not evidence that the definition was implemented correctly and selecting a value that is similar to published values would reiterate the omission subsection 2.1.6 identifies across the literature review.

The Full cohort rates are calculated from the feature matrices as shown in Figure 3.3, this accounts for the SW exclusion of 6,501 admissions that were planned-surgery (subsection 3.1.4) and the removal of 229 positives under the *old* label, which is the only label whose count changes on that basis. The values in Table 3.1 are all computed on the same admissions with the same set of features. Most of the difference under the *old* and *lat* labels in the ICU are due to the same tier movements which account for 21.9% of ICU positives and similarity for SW which have 34.8% positives under *old*.

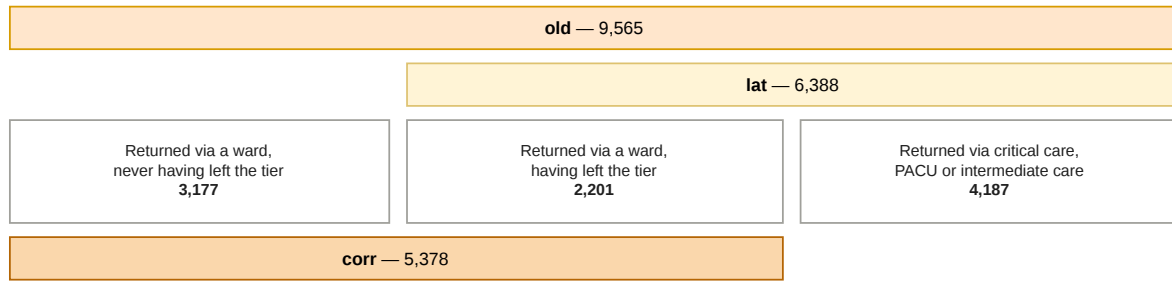


Figure 3.4: Three SW label variants on the feature matrix of Figure 3.3.

Figure 3.4 also exposed an issue with the *corr* label. The route was being assigned by a residual branch which made the *via_ward* mean *returned* and not via *ICU*, *intermediate care*, or *PACU*. A total of 59% admissions of *corr*'s positive class had inter-tier movement, which is the opposite of what the label was created from and this made the label indicate opposite to what it was created for. This is a construct validity failure that subsection 2.1.5 had described in which the rule is applied correctly but the defect lies in what it records. This made an even stronger case for selecting *lat* as the main label for this project. A similar construct validity issue occurred for the PACU unit name and therefore *corrpacu* label applies the correction.

The multiple label variant are examples of a robustness check and not a validation of whether the labels are capturing the intended clinical event.

3.1.6 Independent variables

A total of thirty-nine features are calculated for each admission in three separate groups, which are listed in full in Appendix C.2. After one-hot encoding of the eight categorical features, the features expanded to a total of 76 model inputs and the feature matrix has 42 columns, which are the 39 features with the *hadm_id*, *subject_id* and the label.

Two process features carry no variance at all, them being the *num_icu_stays* which are 85,244 of the 85,248 ICU admissions and the *num_sw_stays* which comes at a total of 67,061 of the 67,066 SW admissions. These two features have no variance at all because the prediction point is at the end of the first stay and the patient has essentially completed their first stay in the ICU/SW ward. Both these process features are kept to compare the difference between the source paper. BMI is missing for 42.6% of ICU and 32.8% of SW admissions, and these values are substituted as an explicit *UNKNOWN* category than imputing values. The ED flag is derived from an earlier version of the event log, which has the same values as the current event log but with a difference of eight admissions in total.

From the 39 feature set, 23 are only available after the prediction point, which has 20 discharge-assigned diagnostic codings (18 ICD chapter counts, Charlson index and

chronic disease count) taken from the *diagnoses_id* column, which are entered at discharge and have no timestamped pre-discharge equivalent. The remaining three are whole-admission aggregates (abnormal-laboratory percentage, medication count and the polypharmacy flag), their source rows have a timestamp and are aggregated with no time predicate, so a prefix-bounded version could have been built but was not. This is a design choice rather than a limitation, since the seven process features are prefix-bounded, filtering on arrival before the prediction point and then capping the duration at it. Both types are look-ahead leakages as explained in subsection 2.1.5 and not the double-use failure Hagmann et al. [25] describes.

The ablation study (subsection 3.2.6) removes the 20 discharge assigned coding and leaves the three whole-admission aggregated features, so this ablation study should be seen as removing the discharge assigned codings but not a temporally valid model. All 23 values are present in every reported result and every feature condition of the subsection 3.2.3.

3.1.7 Control-flow representations

Two representations are built from the activity sequence of subsection 3.1.3, which corresponds to the two order-sensitive encodings of subsection 2.2.2, where both are stopped before the prediction point. Appendix C.5 shows their configuration and the resulting dimensions.

However, the two representations differ in what they preserve. The n-gram counts are order-sensitive within the window but not beyond it and the integer sequences preserves the order throughout. Fitting the n-gram feature set on the training partition prevents the n-grams from occurring only in the held-out admissions from entering the feature space.

Most of the sequences have a mean true length of four which is shorter than the 30 event limit, so padding makes up most of the sequence and hence carries no information. The event limit does not remove any events and the longest sequence in the SW prefix was up to 29 events. The initial SW sequences file had 73,559 admissions but the modelled dataset had only 67,061 admissions, so the intersection between both the datasets was taken, which kept only the admission that occurred in both (subsection 3.1.4).

The model uses the events that happen before the transfer being predicted, by doing so this does not include the transfer itself. If the transfer was included in the prefix then this would give the model the information about the outcome it is made to predict (subsection 2.2.2).

3.1.8 Care-intensity mapping

The Intensive Care Society Level of Adult Critical Care [26] assigns each level of care according to the care the patient will require. 34 of the 46 distinct care units in subsection 3.1.3 have an assigned level. These levels provide an ordered classification of the units of care that subsection 2.2.1 compares it with the unit-name classifications. However, administrative data will record where the patient is located and not what level of care the patient requires. This mapping uses the levels of care that a unit can provide against the patient’s needs. This mapping is constructed using a UK standard applied to an US hospital dataset, so it needs clinician validation before using it as a general model. Figure 3.5 gives the assignment and the transfer-row count at each level.

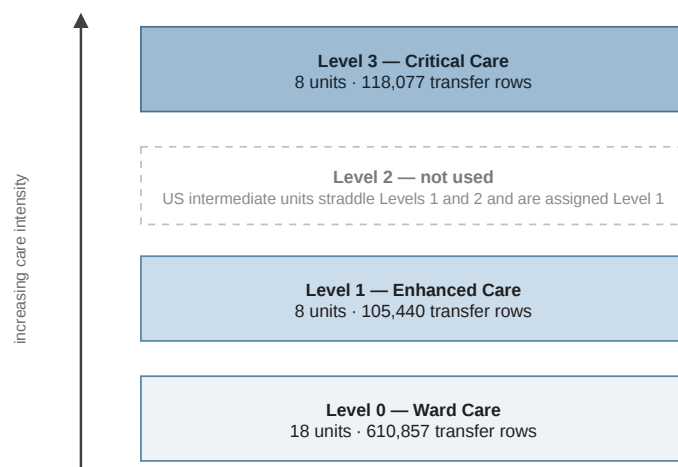


Figure 3.5: Intensive Care Society levels assigned to the 46 `careunit` values. Level 2 is unused because US intermediate units straddle ICS Levels 1 and 2 and are assigned the lower of the two. Transfer-row counts sum to the 2,413,581 rows of Figure 3.1.

The exception is the Post-anaesthesia care unit (PACU), which is assigned Level 1 from the standards but it is excluded as a procedural/recovery flag rather than as a distinct level. Patients record a short stay in the PACU with a median stay time of 2.69 hours, and a total of 61% stay for less than four hours. And 87% of the patients are transferred to it and not admitted to it as a level of care. Assigning the PACU to Level 1 would mean that practically every admission will register a change of level even though the patient’s overall level of care never changed.

This care-intensity mapping and the activity grouping of subsection 3.1.3 are two separate types of classifications and do not classify the units in the same manner. An example to showcase this is that the emergency department and psychiatry are all activities carrying no ICS levels but the Medical/Surgical (Gynaecology) is excluded by the activity grouping but is assigned Level 0 in this mapping.

The mapping is not the primary outcome of this project. It supports Objective 5

(section 1.2) and It provides a method to validate the ward-name labels works as intended. The ward-name labels cannot be used against itself to validate its own findings, so this care-intensity mapping provides a separate comparison between the two. These results are stated in Chapter 5.

3.2 Experimental design

3.2.1 Prediction framework and prefix construction

The prediction framework used here is the outcome-oriented predictive process monitoring (OOPPM), which is a single binary prediction per admission made at a fixed prediction point from the prefix of events preceding it. A total of three decisions subsection 2.2.2 states as preceding the choice of classifier and are as follows:

- Bucketing is settled in subsection 3.1.4
- Encoding is described in subsections 3.1.6 and 3.1.7
- Prefix extraction is explained in this section

No prefix filtering is applied for the traces in the admissions. Each admission yields exactly one prediction point per cohort, so a trace contributes one prefix rather than one per event. The length bias that gave the idea of gap-based filtering in subsection 2.1.2 does not arise here and also as prefix filtering here would remove admission than reduce the redundancy among the prefixes drawn from the same trace. Figure 3.6 explains the entire construction of the prediction framework.

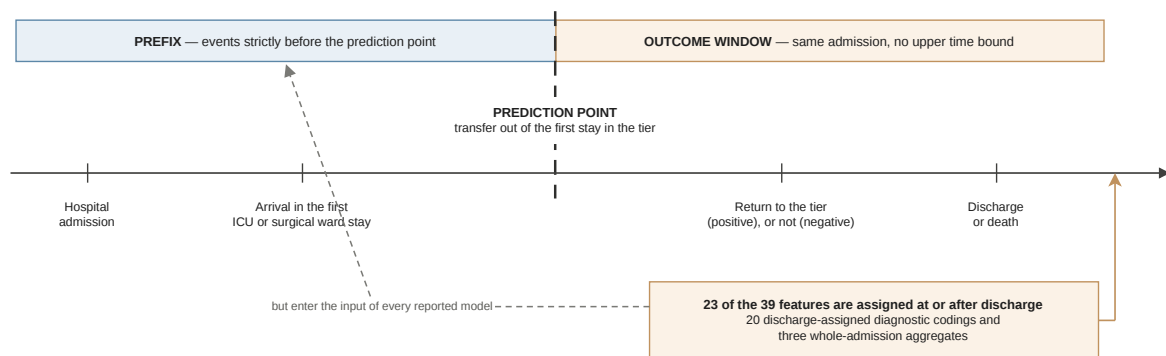


Figure 3.6: The prediction point and the prefix, with each admission having one prediction point per cohort.

3.2.2 Data partitioning

One canonical partition is created per cohort and is used by every model. The admissions are grouped by the *subject_id*, so that all the admissions of a patient fall on one side of the split and are stratified on the outcome. The partition is created by using

nested *StratifiedGroupKfold* with seed 43 and is identified by a *split_id* hash which is recorded on every result row (subsection 3.2.8), this method resulted in zero patient overlap between the splits of the partition.

Table 3.2: Canonical partitions

Cohort	n	Train	Validation	Test	split_id
ICU	85,224	60,666	7,655	16,903	6e09b6e8a1606e84
SW	67,061	47,672	5,952	13,437	0f50fb19f263c430

All models are trained on the training partition, with the validation partition used for early stopping and is never trained on. This makes every model see the same set of rows and it comes from the canonical split rather than being redrawn per model, which would have resulted in early-stop against a different set of patients. Chronological splitting is unavailable in this dataset with the reasoning stated in section 1.4.

The feature matrix and the sequence files do not contain the same set of admissions. A total of 24 ICU and 5 SW admissions are in the feature matrix but not in the sequence files and 6,498 SW admissions appear in the sequence file but not in the feature matrix. Every model uses the intersection of both these files to ensure that the same set of admissions is included and excluded at the same time. The test partitions are also identical across every model and every label variant.

3.2.3 Feature conditions

The tabular models are fitted under three feature conditions and the sequence models under two feature conditions. The *MLP* is the exception which is fitted only under the *No Process* and *With Process*, giving a total of 90 fits rather than 135 like the other models. Table 3.3 lists the conditions with the number of encoded inputs each supplies.

Table 3.3: Feature conditions. Input counts are encoded model inputs, after the one-hot expansion of subsection 3.1.6.

Condition	Inputs	Applies to
No Process	59	Tabular families
With Process	76	Tabular families
With BoW	76 + 280 (ICU) / 268 (SW)	Tabular families
Sequence Only	Integer sequence	Sequence families
Sequence + Tabular	Integer sequence + 76	Sequence families

No Process condition excludes the seven process features of subsection 3.1.6, which contribute a total of 17 of the 76 encoded inputs. *With BoW* appends the n-gram

features of subsection 3.1.7 with the full tabular set, and the *Sequence + Tabular* pairs the integer sequence with the same 76 inputs, so the two sequence condition only differ in whether the tabular features are present or not.

3.2.4 Model families, configuration and repetition

Thirteen models are compared in this project, having both tabular as well as sequence models as defined in subsection 2.2.3. The tabular models take the encoded inputs of subsection 3.2.3, while the sequence models take input the integer sequence of subsection 3.1.7 directly. Table 3.4 groups the thirteen models and gives the number of fitted runs for each.

Table 3.4: Thirteen models, with number of fitted runs.

Group	Models	Runs
Linear	Logistic regression	5
Tree ensembles	Random forest, gradient boosting	5
Distance-based	k -nearest neighbours	5
Feed-forward	MLP	5
Convolutional	CNN ($k = 2, 3, 4$), CNN-k23, CNN-LSTM	5
Recurrent	RNN, LSTM, GRU, BiLSTM, BiGRU	5

Every stochastic model is fitted a total of five times (seeds 42, 1, 2, 3 and 4), and the differences between the five results are used to measure how sensitive the models are to random variations. *Logistic regression* under the *L-BFGS* (limited-memory BFGS) optimisation algorithm and *k-nearest neighbours* return five identical fits and hence their values are reported as a single value instead of a mean with zero spread, as a deterministic model having no variation is different from a stochastic model tests across seeds and returning no spread (subsection 2.2.5). *Gradient Boosting* is counted as stochastic because its splitter permutes features at each node even at *max_features = None*.

Class imbalance is handled by balanced sample weighting in the modes that support it and no resampling is applied such that every model is fitted on the same set of rows/admissions.

Hyperparameter tuning was not prioritised in this project and for the reasons given in section 2.3. The main design effort went into creating the outcome definition, the prefix boundary and the evaluation design. As a result, no model is represented at its best achievable configuration.

Model performance is estimated on a single held-out test set which is kept hidden during the training period. Instead of using cross-validation to estimate variations, the models

were run repeatedly with different seeds and the differences between those runs showed how much the performance varies. The source paper uses a 10-fold cross-validation method. A shared test set is where you can compare between models, since independently drawn test partitions would lead to a difference in performance that would arise from which admissions fell into each fold.

3.2.5 Metrics and operating point

AUROC is the primary metric used to measure the performances as explained in subsection 2.2.5. AUPRC is reported with it against its own no-skill baseline, which equals the test-set positive rate and differs by cohort and also by label variant (0.1188 for ICU *lat*, 0.0941 for SW *lat*, 0.0229 for SW *collapse* and 0.4607 for the restricted SW cohort of subsection 3.2.7). Each AUPRC value is read against the baseline for its own columns.

Precision, recall and F1 score are all computed under two threshold policies. The main model comparison uses a threshold fixed at 0.5 across all the models. The error analysis of subsection 3.2.6 tunes the threshold on the validation partitions to maximise F1 and records the tuned values per run.

The error analysis used the second threshold policy because at a positive rate near 8%, a threshold of 0.5 matches the *Random Forest* predict almost no positives. This results in an empty confusion matrix and leaves nothing to profile against. Tuning is done on the validation partition so that the training partition is kept unseen until the final error analysis. Under either policy, these metrics move with prevalence by how the labels are defined, so these values are not comparable across label variants.

3.2.6 Attribution and error analysis

Attribution addressed Objective 3 by ablation over the feature groups of subsection 3.1.6, under four different views as follows:

- Leave-one-group-out
- Incremental addition
- Single-group-only
- Reduced subset by removing 20 discharge assigned coding, leaving 56 encoded inputs

The fourth view removes the 20 discharge assigned coding rather than creating a temporally correct model, as the three whole-admission aggregates of subsection 3.1.6 still remain in it.

The classifier chosen for the ablation study is *Logistic Regression*, which was held fixed across all conditions so whatever varies is the feature sets and not in the model capacity.

It is deterministic under the *LBFGS*, so one fit per configuration is sufficient and no variance is reported. There are two limitations of this approach as the attribution is conditional on the model selected and the correlated feature groups do not decompose, so their individual contributions cannot be separated cleanly.

Error analysis addresses Objective 4 in which the test cases are tagged by the confusion-matrix cells and then the feature distributions are compared across the cells. Also multiple profiles of different subgroups are created based on the characteristics of the cohorts. Tagging the test cases needs an operating point, and that threshold is selected on the validation partition to make sure that it maximises the F1 score, with those selected values recorded per run (subsection 3.2.5). The values ranged from 0.32 to 0.42 across both the cohorts and the label variants.

The classifier used for error analysis was tuned *Random Forest* at seed 42 and fitted once. As Random Forest is a stochastic model, the subgroup miss rates will carry sampling variation which is not quantified here. The analysis characterised which patients were missed and not which subgroup is significant when compared with the other subgroups.

3.2.7 Robustness designs: label sensitivity and restricted cohort

The label sensitivity analysis re-runs the entire pipeline under each of the five label variants of subsection 3.1.5, whilst keeping the dataset, features, partitions and the model fixed throughout. This would make sure that any changes in the results is attributed to the change in label definition alone. The time parameter is constant across the five, so the analysis tests against the sensitivity of what counts as a return and not to the time window in which it occurs.

The restricted-cohort run excludes the structurally-negative admissions of subsection 3.1.4, which left the admissions in the cohort that could only be a positive outcome. This is altogether a different set of admissions and not a re-sorting of the original admissions set, so hence it has its own partition splits and *split_id*.

Table 3.5: Restricted-cohort partitions, excluding the structurally-negative admissions of subsection 3.1.4. Grouped and stratified as in Table 3.2.

Cohort	n	Train	Val.	Test	split_id	Test prev.
ICU	73,400	52,304	6,402	14,694	14d2082d80407129	14.00 %
SW	14,011	9,946	1,265	2,800	476006eb47fe7cf9	46.07 %

The SW cohort becomes near-balanced at 46.07% compared to the SW 9.41% under the canonical partitions. Class weighting, the AUPRC no-skill baseline and threshold

behaviour all differ as a result, so AUPRC from this run is not comparable with the canonical split and hence reported separately.

3.2.8 Computational environment and reproducibility

The tabular models and all analyses were run on a local Windows workstation. The following were under *scikit-learn* on CPU:

- *Logistic Regression*
- *Random Forest*
- *Gradient Boosting*
- *k-nearest neighbours*

The following were under *PyTorch* on a NVIDIA GTX 1650 GPU:

- The *MLP*
- Ablation study and error-analysis (subsection 3.2.6)

The remaining eight sequence architectures ran on *Aire*, the University of Leeds HPC cluster. They were all ran through *SLURM* array jobs each allocated 8 CPU cores, 32 GB RAM and one NVIDIA L40S GPU per task, following the requirements noted in subsection 2.2.3 that the sequence models might require accelerated hardware to be practical. Appendix C.6 gives the release versions of each components used in both environments.

The two environments differed by release versions of the tools used (*pandas*, *python*, *scikit-learn*), but the *split_id* validation check passed on all partitions and the event-log reconciliation of Figure 3.1 derives exactly. Therefore there was no indication that the different release versions affected the results.

No *requirements.txt* or *environment.yml* existed while the experiments ran, so the versions in Appendix C.6 were read from both environments after the runs were completed.

Chapter 4

Results of the Empirical Investigation

4.1 Experimental programme and reporting conventions

Every result in this chapter comes from one set of model training experiments. A total of 1,350 model fits were carried out in which 720 deep, 540 classical, and 90 for the *MLP*, which were trained under the *No Process* and *With Process* conditions as explained in subsection 3.2.3. The deep-learning architectures consumed an approximate of 4.20 hours of GPU usage on the *HPC*.

The nine cells are the ICU cohort under *old*, *lat* and *restricted*, and the SW cohorts with those three with the additional *corr*, *corrpacu* and *collapse*.

The results are reported as a mean of the five seeds, and with the standard deviation given to show how much the five different seed runs differ in value. *Logistic Regression* and *k-nearest neighbours* are deterministic, hence both are reported as a single value and not a mean with zero difference. *Gradient Boosting* is stable across the groups where in 10 out of the 27 cohort, variant and conditions group, its result changes marginally across the different seed at most by 0.0002, still the standard deviation is reported for the same. *Random Forest*, the *MLP* and the eight deep sequence architectures are stochastic in nature and their standard deviations are reported along with their values.

When two models are being compared, the test set is a paired 95% t-interval on the per-seed difference (n=5) so that the comparison is made across the same five random seeds. An interval that excludes zero is reported as one model performs better than the other, but if the interval includes zero it is reported as a tie and the two models are described as indistinguishable.

Here the primary performance metric used is AUROC and a secondary metric AUPRC is read against the no-skill baseline for its own cells as defined in subsection 3.2.5, which is the test-set positive rate and ranges from 2.29% to 46.07% across the nine cells. Precision is also computed using the same basis. Section 4.4 tabulated the positive rates on the full modelling matrix, which range between 2.27% to 45.59%.

The *restricted* cohort has its own partitions and its own *split_id* (subsection 3.2.7) and its values are reported as a separate finding and are not comparable with the canonical cohorts.

4.2 Replication of the published baseline

Objective 2 is addressed by fitting the five tabular models under the *No Process* and *With Process* conditions as explained in subsection 3.2.3, which is identical to the two feature sets Aljebreen et al. [1] report AUROC under. Table 4.1 gives the comparison between both the studies.

Table 4.1: Baseline replication against Aljebreen et al. [1] on *lat* label variant and AUROC.

Cohort	Model	Without process features			With process features		
		Pub.	Ours	Δ	Pub.	Ours	Δ
ICU	Logistic regression	0.75	0.7439	−0.006	0.95	0.7792	−0.171
ICU	Gradient boosting	0.75	0.7470	−0.003	0.98	0.8167	−0.163
ICU	Random forest	0.74	0.7256	−0.014	0.98	0.8024	−0.178
ICU	<i>k</i> -nearest neighbours	0.60	0.5803	−0.020	0.89	0.5903	−0.300
ICU	MLP	0.74	0.7440	+0.004	0.97	0.8088	−0.161
SW	Logistic regression	0.80	0.7634	−0.037	0.99	0.8781	−0.112
SW	Gradient boosting	0.82	0.7705	−0.050	0.99	0.9167	−0.073
SW	Random forest	0.80	0.7540	−0.046	0.99	0.9121	−0.078
SW	<i>k</i> -nearest neighbours	0.71	0.6408	−0.069	0.94	0.6967	−0.243
SW	MLP	0.82	0.7689	−0.051	0.99	0.9115	−0.079

The *Without process* features in all the ten cells fall within 0.07 of the source paper’s values and the five ICU cells fall within 0.02, the *MLP* differing by +0.004. The *With Process* features fall short in every cell when compared with the source paper’s values, ranging from 0.073 to 0.300. Every SW cell falls below the published values under both the conditions, which holds across all the five models, and attributes the shortfall either because of the cohort definition or the label definition and not any one model. Under the *old* label the SW still has a shortfall in the *without process* features column that ranges from 0.094 to 0.112. Section 5.4 explains why this difference occurs.

Therefore the two conditions replicate differently and the reason behind the divergence is not because of the models as they were held constant across all conditions. The reason is in how the process features are bounded. This project computed all seven strictly before the prediction point (subsection 3.1.6), when compared to Aljebreen et al., they state that the ICU and SW durations were calculated up to the prediction event, but the stay counts and total events were counted over each admission, naming

total events as one its four most important process features. But a stay count taken over the whole admission encodes the outcome directly, since an ICU left shift is a second ICU stay. Under the prefix boundary used here for the two features, they carry no variance at all which subsection 3.1.6 reports.

The ICU cohort used here is 1.56 times the size of the reported cohort size and the difference is not attributed as discussed in subsection 3.1.4, this makes the comparison not like-for-like on cohort size either. Section 5.4 states what was tested and what still remains unexplained. The SW equivalent is given in Appendix C.7.

4.3 Model families and the value of process features

The thirteen models of subsection 3.2.4 were fitted in all nine cells and Figure 4.1 plots all the thirteen under the *lat* label in both the cohorts.

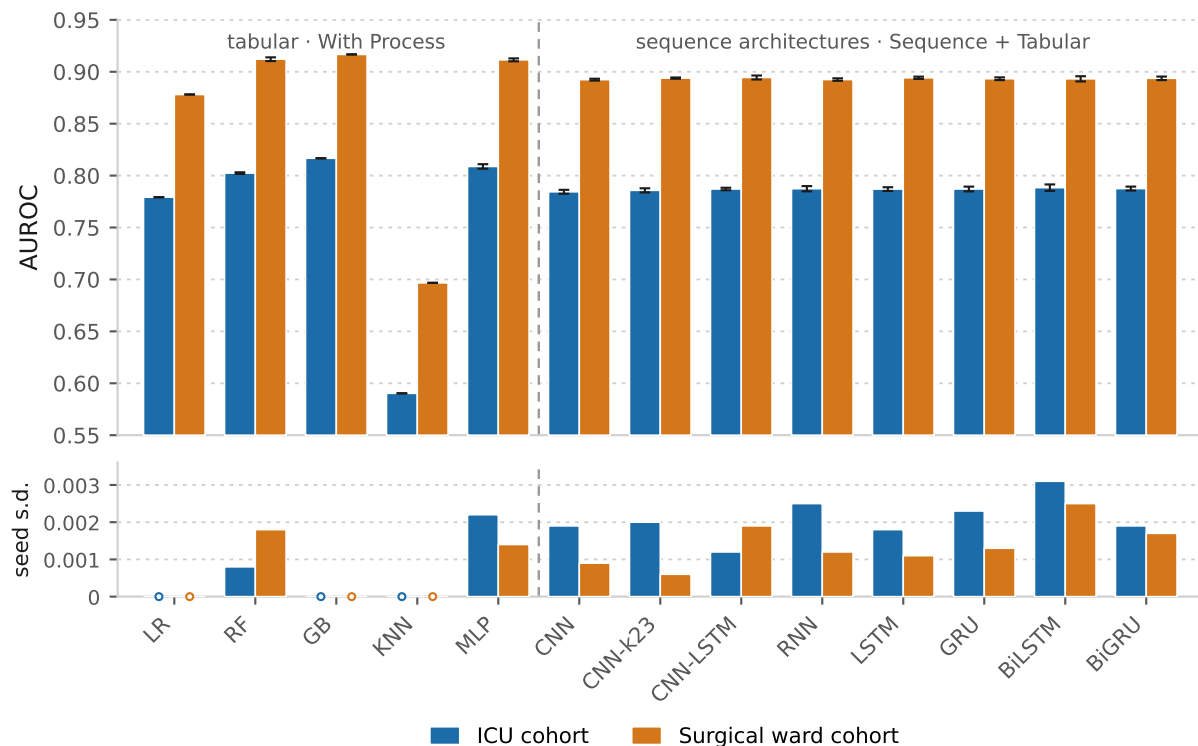


Figure 4.1: Mean AUROC of the thirteen models under the *lat* label variant, both cohorts; the lower panel gives the seed standard deviation on its own scale and the AUROC axis is truncated at 0.55.

Gradient Boosting beats the best neural model in all nine cells, and every paired interval excludes zero as shown in Table 4.2. The smallest margin is in the ICU *restricted* label, where the lower bound of the interval reached +0.0001.

In all the nine cells, the *MLP* is the best neural model, which makes a feed-forward network on the same feature set outperform the convolutional or recurrent models in

every cell. Against the best model in each cell, the difference ranges from +0.0109 to +0.0452 where every interval excludes zero. These two measurements together imply that sequence models have no potential benefit on this specific task at hand.

Table 4.2: Gradient Boosting against the best neural model and sequence model on all nine cells with Δ being the paired per-seed difference.

Cohort	Variant	GB	Comparator	AUROC	Δ	95 % CI
<i>Against the best neural model</i>						
ICU	old	0.8424	MLP	0.8378	+0.0046	[+0.0038, +0.0053]
ICU	lat	0.8167	MLP	0.8088	+0.0079	[+0.0052, +0.0107]
ICU	restricted	0.8097	MLP	0.8081	+0.0016	[+0.0001, +0.0030]
SW	old	0.8986	MLP	0.8948	+0.0038	[+0.0027, +0.0050]
SW	lat	0.9167	MLP	0.9115	+0.0052	[+0.0035, +0.0069]
SW	corr	0.8595	MLP	0.8456	+0.0139	[+0.0114, +0.0164]
SW	corrpacu	0.8577	MLP	0.8482	+0.0095	[+0.0081, +0.0110]
SW	collapse	0.8878	MLP	0.8637	+0.0241	[+0.0204, +0.0278]
SW	restricted	0.8191	MLP	0.8147	+0.0044	[+0.0006, +0.0082]
<i>Against the best sequence architecture</i>						
ICU	old	0.8424	CNN-k23	0.8113	+0.0311	[+0.0290, +0.0332]
ICU	lat	0.8167	BiLSTM	0.7884	+0.0283	[+0.0245, +0.0321]
ICU	restricted	0.8097	GRU	0.7900	+0.0197	[+0.0177, +0.0218]
SW	old	0.8986	CNN-k23	0.8799	+0.0187	[+0.0168, +0.0207]
SW	lat	0.9167	CNN-LSTM	0.8945	+0.0222	[+0.0198, +0.0246]
SW	corr	0.8595	GRU	0.8320	+0.0275	[+0.0254, +0.0297]
SW	corrpacu	0.8577	BiLSTM	0.8323	+0.0254	[+0.0237, +0.0271]
SW	collapse	0.8878	BiLSTM	0.8426	+0.0452	[+0.0420, +0.0484]
SW	restricted	0.8191	LSTM	0.8082	+0.0109	[+0.0085, +0.0134]

Architectural difference is counted as detected when the spread exceeds twice the mean within-model seed standard deviation. Under the *Sequence + Tabular*, the eight architectures are indistinguishable in all the nine cells where the spread ranges from 0.0021 to 0.0062 against the seeds standard deviation of 0.0014 to 0.0033. Under the *Sequence Only*, the difference is detectable in ICU *old*, ICU *lat*, SW *collapse* and SW *restricted* labels. All 32 per-architecture values in these four cells range between 0.5272 and 0.6975, thus the architectural difference occurred within a low-performance range and not across a strong predictive performance.

Sequence-only AUROC on ICU performed only marginally better than random guessing, regardless of which ICU label variant was used. These values were 0.5330 under *old*, 0.5335 under *lat* and 0.5456 on the restricted cohort. This was also similar across the SW *Sequence-only* AUROC where it ranged from 0.5541 to 0.6954. Upon adding the tabular features increased the AUROC by 0.1112 to 0.2771 depending on the cell, and also within one cell the eight deep architectures gained by a standard deviation of at most 0.0042. The gain is almost identical across the eight architectures which implies that the models are benefiting from the tabular information, whilst the sequence information is not of much contribution.

The change from *No Process* to *With Process* across the five tabular model address Objective 3. The addition of the seven process features raises the AUROC by 0.0100 to 0.0768 on the ICU and by 0.0559 to 0.1581 on the SW. *k-nearest neighbours* gains the least from this addition whilst *Random Forest* has the maximum gain reported. Also all the five models report more gain on the SW than on ICU. The *Logistic Regression* gain is the same measurement as the leave-one-out process drop reported in section 4.5, and not a second finding.

4.4 Sensitivity to outcome definition

The gap identified in subsection 2.1.6 is how much the choice of outcome definition alone moves the results within a single cohort. This gap is closed by the five label variants of subsection 3.1.5, in where the cohort, feature set, partition and the model all fixed at the same values and only the label rule changes (subsection 3.2.7). Table 4.3 give the best model per cell under the *With Process* condition.

Table 4.3: Best model per cohort and label variant under the *With Process* condition with AUROC. Prevalence is the positive rate of the full modelling matrix. *restricted* row is on its separate cohort and partition.

Variant	ICU cohort		Surgical ward cohort		Gap
	Prev.	Best	Prev.	Best	
old	14.95 %	GB 0.8424	14.26 %	GB 0.8986	+0.0562
lat	11.68 %	GB 0.8167	9.52 %	GB 0.9167	+0.1000
corr	—	—	8.02 %	GB 0.8595	—
corrpacu	—	—	9.47 %	GB 0.8577	—
collapse	—	—	2.27 %	GB 0.8878	—
restricted	13.57 %	GB 0.8097	45.59 %	RF 0.8199	+0.0102

By varying only the label rules moved the AUROC by 0.0590 in the SW cohort and by 0.0257 in the ICU cohort. This variation in each cohort exceeds the largest margin between *Gradient Boosting* and the best neural model under the *With Process* condition in every cell for that cohort, which is 0.0241 for SW and 0.0079 for ICU (section 4.3). When the model is fixed, the SW difference ranges from 0.0590 for *Gradient Boosting* to 0.1740 for *k-nearest neighbours*. This implies that the outcome definition moves the performance more than the margin that separates the leading models from one another, a claim that subsection 2.1.2 takes from Teinmaa et al. [9] and this study tests this within a single cohort.

The label variants are separate rules entirely rather than different points on a scale. The *lat* label identifies 6,388 SW positives whereas the *corr* identifies 5,378. Both having a

common intersection of 2,201 positive values (subsection 3.1.5). Therefore, the order of the rows in Table 4.3 do not have a specific meaning.

The difference between all five seeds under the *collapse* label averages 0.0081 across the eight sequence model with the *Sequence + Tableau* condition, against the 0.0033 under the *lat* label. The largest difference within a single model across all of the seeds is 0.0112. The range is roughly 2.4 times wider and tracks the 147 validation positives that label variant leaves. It remains below half the *Gradient Boosting* margin in that cell (section 4.3).

The surgical ward advantage

The SW cohort exceeds ICU by 0.0562 under the *old* label and 0.1000 under the *lat* label. However on the restricted cohorts that exclude the structurally negative admission (subsection 3.2.7), the gap becomes 0.0102. These three values are gaps rather than level, which make the restricted cohort comparable here, and they indicate that the gap occurs due to the cohort definition and not on the basis of that the outcomes are easy to predict in the SW cohort. The restricted cohorts removes 79.1% from the SW cohort but only 13.9% from the ICU cohort. However from these exclusions have different causes. At least 81.8% of the SW admission removed were discharged alive from the tier and never returned, but 51.2% of the ICU admissions removed died in the hospital recording no onwards transfer. The breakdown is given in Table 5.3.

Gradient Boosting is the best model overall in every cell except in the restricted SW cohort, where the *Random Forest* scores higher at 0.8199 against 0.8191. The paired difference is +0.0008 with a 95% interval of $[-0.0017, +0.0033]$, which makes the two models indistinguishable in that specific cell. The interval is driven by the *Random Forest*'s standard deviation of 0.0019 against *Gradient Boosting*'s 0.0001.

4.5 Feature attribution and temporal validity

Objective 3 is addressed by ablation over the feature groups, with *Logistic Regression* held fixed across all the conditions stated in subsection 3.2.6. The clinical groups of subsection 3.1.6 are ablated as its four constituents, that are the *clinical core* of the seven features, the eighteen *ICD chapter counts*, the *Charlson index* and the *chronic disease count*, while the *Demographics* and the *Process* features are ablated whole. Table 4.4 gives the leave-one-out view under the *lat* label variant and the single-group and incremental views for both the cohorts are in Appendix C.8.

Table 4.4: Leave-one-out ablation over the six feature groups with AUROC drops.

Group removed	Variables	ICU drop	SW drop
Clinical core	7	0.0703	0.0638
Process	7	0.0353	0.1147
ICD chapters	18	0.0057	0.0016
Chronic disease count	1	0.0008	0.0012
Charlson index	1	0.0000	0.0000
Demographics	5	-0.0002	-0.0001

The two cohorts differ in which features are more important, where the single-group view agrees with the leave-one-out drops. The process features alone reach 0.7715 on the SW cohort and 0.5592 on ICU. Both views place the process features as the dominant group in the SW cohort and the secondary group in ICU, in which the clinical core feature dominates. However the clinical core includes the three whole-admission aggregates of subsection 3.1.6, so its ICU drop is not entirely attributable to the information available at the prediction point.

The five demographic features and the *Charlson index* move at exactly 0.0000 or below in both the cohorts with different reasons. *Demographics* reaches only 0.5190 alone on the ICU cohort, so they carry almost no signal there. The *Charlson index* reaches 0.5845 on ICU and 0.5621 on the SW, and *Demographics* reach 0.6096 on the SW. These groups are not uninformative but the information they carry is already present in the *clinical core* and the *ICD chapter counts*.

As the *process* group is added last, its incremental gain is almost the same as in the *leave-one-out* drop which is +0.0353 and +0.1147. The incremental view keeps the order in which the feature groups were added and not providing their independent evidence. As the groups are correlated (subsection 2.2.4), their individual contribution cannot be attributed directly, so hence the *single-group* and *leave-one-out* values do not sum to the total gain reported.

The cost of removing discharge-assigned coding

The fourth ablation method removes the 20 discharge assigned codings but retains the three whole-admission aggregates as described in subsection 3.1.6, so this makes the last ablation method without the discharge assigned coding but not temporally valid.

The impact of removing these discharged-assigned coding differs in both the cohorts. On the canonical cohorts, the cost is under 0.009 in every cell, with the largest value being in the SW *corrpacu* label variant at 0.0083. Under the *collapse* label, the reduced model scored 0.0028 higher, which is the only cell where removing the coding increases the AUROC.

For the restricted cohorts, the cost rises in both very different amounts. In the ICU, it roughly doubles, from 0.0061 under *lat* to 0.0116. In the SW cohort, it rises more than tenfold, from 0.0035 to 0.0402. This asymmetry indicated that when the structurally-negative admissions are removed, the SW cohorts depends on the information unavailable at the prediction point to a degree that the ICU does not.

4.6 Error analysis

Objective 4 is addressed by tagging the admissions by confusion-matrix cells. Table 4.5 gives the resulting operating points for each group.

The tuned threshold ranges from 0.32 to 0.42, as the fixed threshold of 0.5 the model retained between 0.226 and 0.640 of the positive cases, when compared against the retained from the tuned thresholds came at 0.517 to 0.753. On the ICU cohort under *lat* label, the true positive cell falls from 1,081 admissions to 453, which is a low value to use for the group comparisons in Figure 4.2.

Table 4.5: Operating points for the error analysis

Variant	Cohort	Threshold	n test	Recall	Precision
old	ICU	0.40	16,903	0.590	0.480
lat	ICU	0.36	16,903	0.538	0.375
restricted	ICU	0.38	14,694	0.517	0.424
old	SW	0.40	13,437	0.731	0.482
lat	SW	0.41	13,437	0.583	0.494
corr	SW	0.36	13,437	0.604	0.301
corrpacu	SW	0.32	13,437	0.666	0.328
collapse	SW	0.41	13,437	0.617	0.150
restricted	SW	0.42	2,800	0.753	0.704

Precision is influenced more by prevalence than by the model’s ability to discriminate between outcomes. For the *collapse* label, at a test-set prevalence of 2.29% the precision is 0.150. In the restricted SW cohort it has a prevalence of 46.07% and precision of 0.704, which is on the same pipeline with a lower AUROC value, when compared between both the cohorts it is 0.8254 against 0.8810.

Missed positives

In the ICU cohort, the false negative admission carries fewer medications, fewer abnormal laboratory results, a lower Charlson index and fewer chronic conditions than the true positives, and also those admitted stayed longer in the unit. In the SW, the false negative carries more accumulated duration, more prior ICU durations, more recorded events, and more prior ICU stays and unplanned-return events, at 0.393

against 0.041 and 0.759 against 0.168. This difference shows that the ICU model missed those patients whose recorded severity does not flag them, and the SW model missed those patients who have a difficult prior critical-care course.

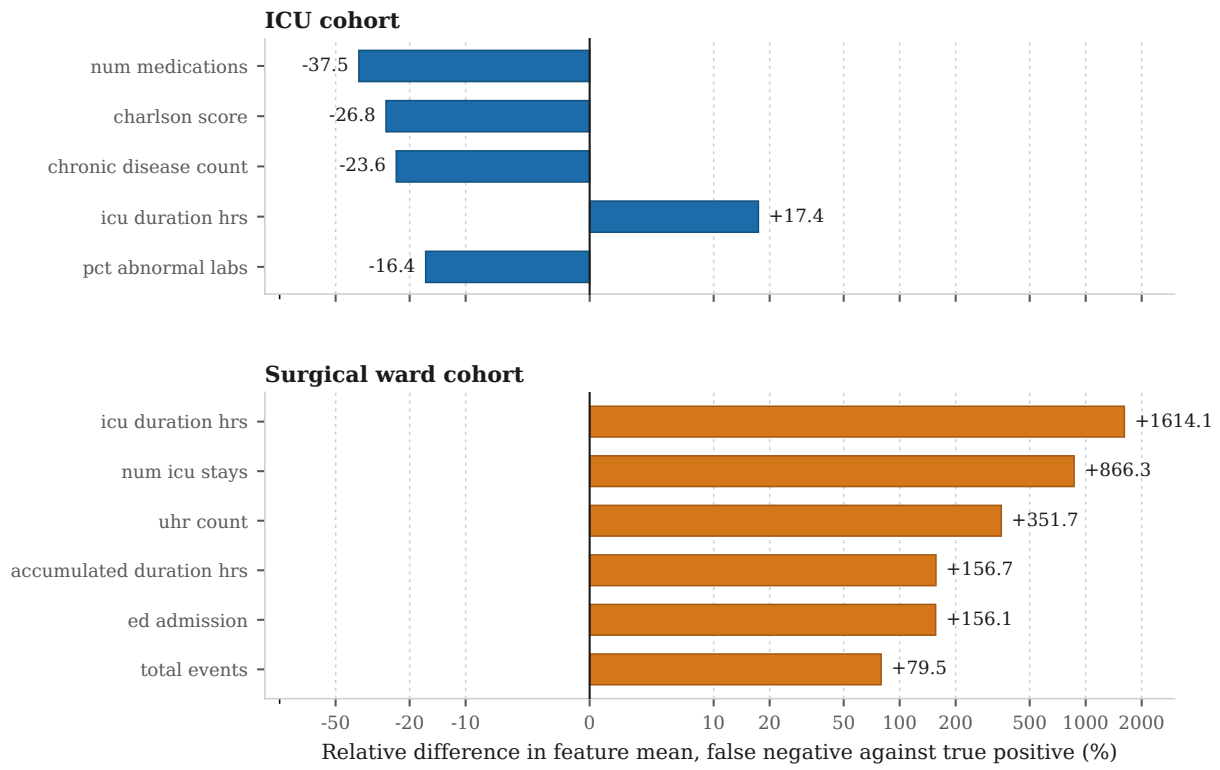


Figure 4.2: Feature means in the FN cell against the TP cell

A miss rate is the proportion of positive admission in a subgroup that are missed by the models. The miss rate in ICU is 0.462 and 0.417 in the SW under the *lat* label, and it also varies across the 53 subgroup cells which satisfy the 20-positive threshold. This variation is also different in both the cohorts where the SW variation is larger than the ICU for five of the six characteristics. The largest SW variations are by age band (0.316 to 0.702) and by named-group ethnicity groups (0.391 to 0.651). When compared with the ICU cohort, the age band variation is from 0.421 to 0.661 and in the named-group ethnicity group is 0.447 to 0.516. The gender groups were an exception in both the cohorts, where the two rates differ by 0.013 in the SW and by 0.032 in ICU.

Four subgroups had fewer than 20 positive cases, so the results from those groups were considered unreliable and were excluded in the analysis. They were the ICU Multiple and Native ethnicity, SW Native ethnicity, and the SW underweight BMI. Out of the four here, three of them are the smallest ethnic groups, hence the removal of them matters with respect to the concerns stated in subsection 1.4.3. It is not known that the observed differences in these subgroups are genuine or just a random sampling variation as a stochastic model is fitted once for this analysis and the confidence intervals were not calculated for these subgroup results (subsection 3.2.6).

4.7 Care-intensity levels

Objective 5 is addressed by the care-intensity scale as described in subsection 3.1.8. This mapping assigns each care unit a level and ordered them according to a 3-level system where the levels are ward care, enhanced care and critical care. This mapping runs over 73,315 admission of the SW extract of subsection 3.1.4 (73,567 minus 252 obstetric, neonatal and psychiatric admissions), which keeps the 6,501 planned-surgery admissions that the main modelling cohort excludes. So both the cohorts used in these two experiments are a subset of one whole dataset, having 66,814 admission in common. The mapping cohort that has 73,315 admissions yields a total of 104,228 care-level segments and the consecutive stays at the same level being merged into one.

A re-escalation is this mapping's implementation of left shifts as described in subsection 1.1.3, which in this context is when a return to a higher level of care occurs after a descent. From the mapping cohort, a total of 2,226 admissions re-escalate to a higher care unit which accounts for only 3.04% of the mapping cohort. Also in that cohort, 64,556 traces (88.1%) have fewer than a three care-level segment and cannot qualify, so the remaining admission come to 8,759 and the detection rate for them comes at 25.41%.

The initial descent to ward care occurs in 2,160 of the 2,226 detections, then the admission that reach critical care are 2,013 (90.4%) and a total of 1,947 (87.5%) escalate directly from ward care into critical care skipping the enhanced care level completely. This escalation detected is therefore sharp rather than gradual increase.

The initial implementation of this mapping placed the emergency department below ward care, but under the Intensive Care Society scale, the ED is a location in which any level of care may be delivered rather than itself, so the final mapping excludes the ED outside the ordering. The initial implementation detected 3,811 re-escalations against the current mappings 2,226. This difference changed 1,589 admissions from positive to negative, with almost all involving an emergency department event.

The mapping is built on a separate SW extract, so it only reaches ICU admissions where the two cohorts overlap, which is 16,965 admissions, 56 less than the shared 17,021 admissions between the cohorts. Whether the ward-name label variants of subsection 3.1.5 agree with the care-intensity mapping is more about the outcome definition and not the methods used, and it is reported in Chapter 5.

Chapter 5

Validation of Results

5.1 Validation strategy

Validation is needed at several parts of this project, with each part addressing a different question. The data part validation is whether the event log actually represents what is in the original data (section 5.2). The partition part is if each model used the correct training and test partition, meaning that the models were trained and evaluated on the data that they were assigned to (section 5.3). Now for the external validation is that does these results from this project reproduce the baseline source (section 5.4). Another validation strategy would be that does the label selected actually measures the clinical event it is made for (section 5.5). The robustness validation is where if changing the prediction definition or usage of a different event log changes the results found (section 5.6). In section 5.7 it states that even after all these validation checks, there are some aspects of the project that cannot be validated against.

Subsection 2.2.6 stated three approaches to validating a label generated from transfer data. The three approaches were the following:

1. Chart review and clinician opinion
2. Correlation with downstream outcomes after prediction point
3. Internal consistency between the defined rule and developed code

Out of the three, only one was validated throughout. Chart review and clinician opinion validation were not possible as clinical time was not available. The label was not also checked against the later outcomes due to time constraints.

The only validation of the label was done through checking whether the definition matched the developed code. In section 5.2 it states that the labels were re-derived from the transfers log using a separate implementation. This tested that whether the stored labels and the new re-derived labels matches. If there is an exact replication of the labels, then this would mean that the stored labels are correctly implemented. However in section 5.5, it points out that even if the rule is correct, the category it produces may not actually describe what it selects. So therefore, even if you re-run the rule on a separate implementation, if the original rule is wrong in its meaning, the re-run will in turn produce the same results.

There are two possible problems with a label: the rule implemented incorrectly, or the

rule is implemented correctly, but it actually measures the wrong clinical activity. To address these problems, this project introduced a care-intensity mapping (subsection 3.1.8), to compare the care-intensity categories with the ward-name labels in section 5.5. The mapping is validated against an external published standard so that the levels make clinical sense. This validation approach was not mentioned in subsection 2.2.6 as it was used as an additional check on top of the three mentioned.

5.2 Data-level validation

Every row from the transfers table were accounted for while constructing the event logs. The event log derivation is shown in Figure 3.1. The three exclusions and the synthetic admission anchor all account for the 1,935,072 events in the event log. There were no residual rows that were unaccounted for, and hence no data was lost during the construction.

The label rule, as defined in subsection 3.1.5 was taken and re-computed on both the ICU and SW cohorts. The new script was created, and the original transfers table was passed through it, and it gave an exact 100% agreement on the stored labels against the new generated labels meaning all the 85,248 ICU and 73,567 SW admissions were re-derived correctly. This re-computation was carried out by using a separate script written after the original label script was made, providing evidence that the stored labels correctly follow the defined label rule.

However this recomputation starts from the same intermediate data the original script read from. So this validates the label and cohort-membership across the intermediate data but cannot establish that the intermediate data was extracted correctly from the original source data.

The modelled cohorts only includes the admissions that appear in both the feature matrix and the sequence files (subsection 3.1.4). In total there are 24 ICU and 5 SW admissions that have features but no entry in the sequence files; hence, these are dropped from the cohorts. After removing those, the modelled cohorts contain a total of 85,225 ICU and 67,061 SW admissions. Now in these modelled cohorts, 11,824 ICU (13.87%) and 53,050 SW (79.11%) admission are all structurally negative. These structurally negative and onward-transfer cases sum up exactly to these totals above, so the cohort counts are internally consistent.

5.3 Partition integrity and the provenance gate

Every partition (train, validation and test split) file gets a unique `split_id`. It is created from the *admission IDs* in each split using SHA-256 (Secure Hash Algorithm 256-bit), so

each split gets its unique value. When the models are fitted, they record which *split_id* it used and is attached with the results. The *collect_results.py* script checks the *split_id* before merging the results. If the results from the same cohorts in a different variant group have different *split_ids*, it blocks the merge and flags it. The four expected *split_id*'s are listed in Table 5.1 and the partition sizes are given in Tables 3.2 and 3.5.

Table 5.1: Partition digest with *split_id*

Partition	Cohort	split_id
Canonical	ICU	6e09b6e8a1606e84
Canonical	Surgical ward	0f50fb19f263c430
Restricted	ICU	14d2082d80407129
Restricted	Surgical ward	476006eb47fe7cf9

Patient IDs were also verified directly at every partition boundary and found zero patient-level overlap. It was not assumed that the *StratifiedGroupKFold* had prevented the overlap and hence it was verified directly.

The shell script files that commands the HPC to run the models and selects the partition per variant. However due to a scripting error in the restricted cohort runs, the shell script was pointing towards the canonical partition instead of the correct restricted partition. The runs were still completed successfully and produced plausible results, so this error was not visible from the outputs alone. When the results were merging, it flagged that the restricted results contained two different *split_ids*. A total of 240 deep results files were created, out of those 40 files were affected by the error containing 160 of the 720 deep model fits. The remaining 200 files were not affected and were confirmed to be correct. The error was resolved and the 40 runs were repeated and then the merge passed in all four groups.

The partition error affected only the sequence-only models results in the two restricted cells because the script launches the eight sequence-model architectures on the HPC. The tabular models and MLP were ran locally and were not affected. The 40 incorrect files affected the specific sequence-model results, while the *random forest* and *gradient boosting* results that were used to compare the two restricted cohorts was unaffected.

Defects the gate did not cover

The ablation and error-analysis scripts that were supposed to use different label variant (*lat* and *restricted*), hardcoded the ICU feature matrix and ignored the environment variable that selects the label variant. This resulted the ICU results labelled as *lat* or *restricted* generated using the wrong data configuration. This script error was found on 24 August because the outputs for the different variants were byte-for-byte identical. The scripting error did not affect the SW results. Every ablation and error-analysis

value reported in section 4.5 and section 4.6, were computed again so the reported figures came from the corrected runs.

The care-intensity results was produced using an older version of the mapping, but the results file did not specify which version it used. The file was updated to use a new version of the mapping scale, meaning the old result file was based on the previous version. Section 4.7 explains the correlation it had on the results.

5.4 Reproduction of published results

Aljebreen et al. [1] reported AUROC values for *logistic regression*, *gradient boosting*, *random forest* and *k-nearest neighbours* with other model types as well. Table 4.1 gives the comparison of the ICU *No Process* condition with the above four. For the *lat* variant, the results were within 0.02 AUROC of the reported values and the *old* variant was within 0.011 AUROC. The differences were marginal and these results were considered as a reasonable reproduction of the published results from the ICU baseline.

The SW results however do not replicate the published results (subsection 4.2). For all four model types, the SW AUROC is lower than the reported values of Aljebreen et al. [1] there was no attempt of diagnosis of this difference as the cause could not be established from the project’s evidence. The *old* label rule has a large difference that ranges from 0.094 to 0.112 AUROC below the source. When using the *lat* label, which by definition is similar to the prediction point described in the source, the gap ranges from 0.037 to 0.069. This decrease in range could suggest that the way the outcome label is defined it explains the short-fall of the values when compared.

The cohort sizes also differ in size with the cohort sizes reported in the source paper. The SW cohort is reasonably close to the reported size being only 1.09 times greater than the source paper. The ICU cohort is significantly larger than the reported size being 1.56 times the source paper’s size. This large gap cannot be reconciled using the available information from the source. Two possible reasons could be accounted for the gap were tested but failed. Removing the coronary care unit from this project’s ICU cohort left with 76,505 admission, which is still greater than the reported 54,782 admission in the source. The source paper described certain limits on the length of patient traces, hence applying the same here removed only 9 ICU and 6 SW admission. However the source explicitly stated that prefix filtering was not applied. So the gap in cohort sizes remains uncertain.

The MIMIC version Aljebreen et al. used cannot be identified, as it is not mentioned in the paper. The source only stated that it used MIMIC-IV covering more than 380,000 patients between 2008 and 2019. When cross-referencing MIMIC’s release notes it was still unclear which version it applied. Also the paper states that it removed cases with

obvious data-quality issues, but does not state the exact issues faced. Therefore the original cohort from the source cannot be fully replicated.

Metric agreement and cohort agreement are two different aspects and this project achieved each in a different cohort. The ICU cohort reproduced the source paper’s metrics but the cohort size was substantially larger than the source. The SW cohort matched the reported cohort size but the metrics fell short. Therefore, proximity to AUROC does not mean that the same definitions were applied and the same is argued in subsection 2.1.6. The cohort sizes were not fully replicated as the dataset version and exclusion criteria could not be reconstructed properly.

5.5 Construct validity of the outcome definition

This project ran two validity checks on the outcome definition, and both these methods are described in subsection 2.2.6. The re-calculation of section 5.2 cover the replication of the source paper’s label rule stated in subsection 3.1.5, and it was successfully replicated to a 100% label match between the original label rule and the reproduced labels. The other validity check was to look at the route assignment and compare it with the name of the category. It flagged a problem with the *corr* label. The *via_ward* is a residual category as it catches the cases that do not fall into the other route categories. As a result, 3,177 out of the 5,378 admission, *corr* marked them positive but they never left the care tier (subsection 3.1.5). The same issue occurred in *corrpacu* as it was inheriting the same residual category. The *collapse* label merges consecutive rows in the same care unit before applying the rule, which removes all of it, at 58 out of 1,520 positives, reducing the problem from the 3,177 out of 5,378. Re-computation could not have found this defect, as the rule was being implemented as it was described. Subsection 2.1.5 documents this failure mode from published literature and it is found in this project’s own labelling.

Changing the label definition varies the outcome (section 5.6), however changing the rule and checking the results change does not indicate that if the rule is actually representing the correct clinical event or not. To validate those results, it would require the chart review and clinical adjudication as mentioned in section 2.3 as unobtainable.

The Care Intensity Mapping against the ward name labels

Objective 5 states whether the ward-name outcome behaves as its own rule’s definition applies, and it cannot test against itself (subsection 3.1.8). The care-intensity mapping provides an independent way to judge the labels as it made from a published external standard. This mapping gives the care categories an ordered scale, so the categories can be compared in terms of care intensity. The eleven activity categories are an unordered set and do not describe which categories represent higher or lower care than the other.

Table 5.2 shows the agreement between the ward-name labels and this external care-intensity mapping.

Table 5.2: Agreement between care-intensity re-escalations and ward-name label variants, on the common admissions. Jaccard is intersection over union of the two positive classes. Share is the fraction of ward-name positives that are re-escalations.

Cohort	Variant	n	Positives			Jaccard	Share
			Ward-name	Re-esc.	Both		
ICU	lat	16,965	2,333	2,128	1,903	74.4 %	81.6 %
ICU	old	16,965	2,964	2,128	1,996	64.5 %	67.3 %
Surgical ward	lat	66,814	6,353	2,226	1,084	14.5 %	17.1 %
Surgical ward	collapse	66,814	1,500	2,226	36	1.0 %	2.4 %

Under the *lat* label, 81.6% of the ICU left shifts are also classified as care-intensity increases whereas the 17.1% of the SW left shifts when compared with the mapping. This difference is structural as the Intensive Care Society scale say that SW and medical wards have the same care level, so this make a return to the SW a lateral move and the mapping scores it zero.

The ICU and SW cohort behave differently when compared with the care-intensity mapping. The ICU outcome generally represents a move to a higher level of care whereas the SW outcome does not. It is more like an unplanned return to the SW rather than an escalation in the care intensity. Aljebreen et al lists unplanned ICU admissions, unplanned SW admission and unplanned re-operation as three separate keywords under the left-shift definition. So this states that treating ICU and SW outcomes as the same definition may be misleading. This conclusion is entirely based on this particular care-intensity mapping being used and not an absolute clinical fact.

The care-intensity mapping was created from the SW data and not independently constructed from the ICU data (section 4.7). A total of 16,965 ICU admissions are shared with the SW data, so the mapping is being evaluated directly on one cohort and indirectly on the other.

5.6 Robustness of the findings

The robustness validation experiment kept the cohort, features, partition and the model fixed and only varied the label rule (subsection 3.2.7). The ICU has 2 label variants and SW has 5 variants in which the *lat*, *corrpacu* and *collapse* variant only change the SW label rule only. With the restricted cohort, it produces nine cohort and label rule

comparison cells. Gradient boosting beats the best neural model in all nine cells and the MLP is the strongest neural model in all nine (section 4.3). The AUROC values do not hold the same way. Upon changing the label rule, it has a bigger effect on performance than changing the classifier within a cohort. SW label-rule difference is 0.0590 and the largest classifier difference is 0.0241 and for ICU 0.02567 and 0.0079 respectively (subsection 4.4). So the general model family conclusion stays the same but the performance changes substantially depending on how the outcome is defined.

The restricted cohort

A separate robustness test where the cohort changes rather than changing the label rule. The restricted cohort removes the admissions that cannot possibly become positive under the outcome definition. After the removal, new train/validation/test partitions are created and each get their own split_id (subsection 3.2.7) so that it does not use the original cohort's partition by mistake. This changes the cohort population rather than the rule.

The two cohorts lose different population and are described in the Table 5.3 below.

Table 5.3: Discharge outcome of structurally-negative group against group with onwads transfer. Each row is a percentage of its own group.

Cohort	Group	Died	Hospice	Alive	Not recorded
ICU	Onward transfer	5.13 %	2.37 %	91.92 %	0.57 %
ICU	Structurally negative	51.22 %	3.01 %	42.75 %	3.02 %
Surgical ward	Onward transfer	3.95 %	1.73 %	88.84 %	5.47 %
Surgical ward	Structurally negative	0.41 %	0.33 %	83.75 %	15.50 %

In the ICU cohort, 54.2% of the excluded admissions ended in death or transfers to end-of-life care, as these patients could not produce the positive outcome. In the SW cohort, only 0.7% of the excluded admissions ended in death or end-of-life care, instead 83.75% were discharged alive. The exclusion removed patients that survive, this means the SW cohort contains many patients whose eventual outcome was already effectively determined before the prediction was made.

The discharge location is not recorded more often in the excluded group of both the cohorts. In the SW, discharge location are missing in 15.50% of the excluded group compared to the 5.47% of the included group. And in ICU, it is 3.02% compared to 0.57% respectively. The missing discharge information is more common among the excluded admissions in both cohorts and not associated with one cohort specifically.

The ICU percentage is lower compared to SW, but the increase relative to its included group is larger. An edge case was tested where the most favourable possible assumption for the SW and assume every missing discharge as a death, while assuming none of the missing ICU cases was a death. The SW reached 15.9% against the ICU cohorts 51.22%. Even after this edge case, the SW figures would be still far below the ICU figure.

The gap between the cohorts falls from +0.1000 under *lat* to +0.0102 on the restricted cohort (section 4.4). This suggested that the advantage of the canonical SW result does not remain once the population whose outcome was already fixed before the prediction point is removed as measured in Table 5.3. The two restricted cohorts are not comparable with the canonical cohorts as both of them contain different populations. In the modelling matrix, the restricted SW cohort has a balanced 45.59% positives compared to only 9.52% in the canonical cohort. This change in class balance affected other metrics and model behaviour as class weighting is calculated using the training data, AUPRC's no-skill baseline depends on the test-set prevalence and classification thresholds behave differently.

5.7 Threats to validity

The care-intensity mapping that the ICU and SW outcomes represent different types of events (section 5.5). If the outcomes are valid, then the comparison in section 4.4 may not be a correct comparison between the two cohorts doing the same prediction task, and hence this makes section 5.6's cohort-comparison explanation incomplete and not incorrect. This conclusion is entirely based on this particular care-intensity mapping being used and not an absolute clinical fact.

No model had its hyper-parameters searched to find its best settings. *Logistic regression* and *Random forest* used balanced class weights, and *logistic regression* also had higher iteration limit. The *MLP* and the eight sequence models used the same manually chosen settings (64 hidden units, dropout = 0.3, batch size = 256, Adam (Adaptive Moment Estimation) learning rate = 0.001 and early stopping based on validation AUROC with patience at 10). The sequence models additionally use an embedding, where the *MLP* does not need. The comparison is not between fully tuned versions of each model and it is more of a comparison using mostly default or manually fixed settings. A tuned neural model might perform better and reduce the observed gap. *Random forest* was used for the error-analysis, and for that it was manually configured with 200 trees and a maximum depth of 20. This model configuration was only used to generate the confusion matrix and not reported as part of the model comparisons.

A total of 39 feature-sets, of which 23 are not available at the prediction point (subsection 3.1.6). 20 are diagnosis codes assigned at discharge and the other three are

calculated from the whole admission, so they can include information from after the prediction point. The ablation removes the 20 discharge features but keep the three whole admission features, which leaves the analysis visible to some future information. Upon removing the 20 features it has a relatively small impact in the canonical experiments (drop of 0.009 AUROC). However the reduction is larger in the restricted cohorts. 0.0402 in the SW and 0.0116 in the ICU (section 4.5).

The error analysis is a single fit at seed 42 and on *random forest* which is stochastic. This only describes the observed errors and does not prove a strong statistical evidence. The model's miss rates are not same across the 53 demographic subgroup cells that had at least 20 positive cases. Some groups have much higher or lower miss rates than others and especially across age groups in the SW cohort. The ethnic group for SW ranges from 0.3901 to 0.651 and for ICU from 0.447 to 0.516. The age groups miss rate ranges from 0.316 to 0.702. As the model was fitted only once, you cannot distinguish whether the observed differences are real differences between groups or simply a sampling variation.

Chapter 6

Conclusions and Future Work

6.1 Conclusions

Across all nine cohort and variant cells, changing how the outcome was defined drove the predictive performance more than changing the model. The apparent difference between the two cohorts reduced once the cohort was restricted. An overall observation was that the models behaved consistently but the definitions underneath them did not.

What the study establishes

The feature pipeline, the attribution method and the error analysis are developed successfully (Objectives 1, 3 and 4). The attribution analysis showed that different feature groups dominate in the two cohorts. The process features carry more influence in the SW whereas the clinical core carries more influence in the ICU cohort. The error analysis found two opposite ways the model was failing, where the SW missed the patients with difficult previous critical-care courses and the ICU missed the patients whose recorded features do not indicate their actual severity (sections 4.5 and 4.6). Also the error rates are not consistent across the different demographic subgroups.

Replication in the ICU baseline was reproduced closely and in the SW baseline was reproduced in partial agreement (Objective 2). All four classical models fall within 0.02 AUROC of the ICU baseline, but the SW fell short against the baseline. The ICU cohort is 1.56 times the source paper's cohort size with the difference unattributed (section 5.4).

The outcome definition was more influential on predictive performance than the choice of model under these tested conditions. Across the same admissions, when the label rule was changed, it moved the AUROC of SW by 0.0590 and for ICU by 0.0257. By comparison, across the same admissions, changing from the best neural model to *gradient boosting* moved the AUROC by a smaller difference, ranging from 0.0016 to 0.0241 (sections 4.3 and 4.4). The eight sequence architecture produce indistinguishable results once the tabular feature are included, the reason being that the average sequence length was 3.4 and in order to maximise the capabilities of sequence models, it needs lengthier sequences. The *MLP* was the best performing neural network across all the tested cells. An AUROC depends heavily on the outcome definition and not the choice of a model.

The same conditions holds for the cohorts as well. The SW advantage drops from

+0.1000 to +0.0102 once the admissions that cannot produce a positive outcome are removed (section 5.6). The care-intensity mapping built for Objective 5 has 81.6% of ICU left shifts as re-escalations, compared to the 17.1% of SW left shifts (section 5.6). The two outcomes may therefore not represent the same underlying clinical construct.

What it does not establish

Validation of the labels to ensure the clinical event it names could not be obtained without a chart review or clinical adjudication. The models were evaluated in offline mode and not in a real-world environment. So it cannot be claimed that these reported performance would hold in deployment. The models were not hyperparameter-tuned, hence the models were not compared at their maximum achievable performance. The ICU cohort is 1.56 times the source paper’s size, but the available evidence is not sufficient to deduce the large difference in value. The care-intensity mapping indicated that the ward-labels behave differently, but this conclusion is on the fact of this particular mapping and not a definitive clinical statement.

Contribution Predictive performance in this setting cannot be solely interpreted by the architectures selected but also how cohorts, outcome, features and evaluation methods are defined. This is demonstrated through the replication gap, the analysis of the cohort effect, and the provenance mechanism that caught a silent partition defect, with an explicit account of where the validation checks do not apply and what they failed to detect.

6.2 Future work

A prefix-bounded feature set, and true cost of temporal validity.

A different approach could be used to rebuild the feature pipeline to include the three out of the twenty-three unavailable features at prediction point. The three whole-admission (abnormal-laboratory percentage, medication count and polypharmacy) have a pre-discharge equivalent and their source rows carry timestamps and are aggregated with no time predicate making it convenient to prefix-bound the feature set. The other 20 features are discharge assigned diagnostic codes with no pre-discharge equivalent, hence they cannot be included. By prefix-bounded feature set, it would give the first temporally valid figures and it would validate the reported values in this project.

Clinical adjudication of the labels.

This project has five label variants and they are not validated against the clinical event it names to predict. A small subset of admissions could be reviewed by a clinician against each rule and it could help establish how well each definition identifies the genuine deterioration. The main obstacle here is the clinical time and not the computing

resources needed, which is why this project could not get the labels validated. A clinical adjudication could also help identify whether the *corr* route-category defect was an implementation error or an issue of the ward-naming labels.

A patient-level care-intensity mapping

The ICS mapping used in the developed unit-capability care-intensity above, describes the capability of a unit and not the actual level of treatment an individual patient receives. This creates a limitation because two patients in the same unit may receive very different levels of treatment. This would mean to use actual treatment information such as vasopressor use, renal replacement or ventilation usage to help identify patients receiving different levels of intensive care within the same care unit. This would test whether the left-shift count moves once the mappings reflects the treatment instead of the location. However, this approach can only be applied to ICU wards as general wards do not report such activities.

Publication of feature-attribution results.

The ablation methods of Objective 3 produced one finding that are independent on how the labels are defined. Different features dominate in both the cohorts, in the SW cohort the process features carry more signal at 0.1147 while in the ICU cohort the clinical core carry more signal at 0.0703. This is a possible finding about the predictive signal that sits in the transfers data, and it is an extension of this project. In order to extend this, the features need to be made fully temporally valid, the models fine-tuned, and the analysis to be replicated on another dataset to identify if the same pattern appears elsewhere.

References

- [1] Abdulaziz Aljebreen, Allan Pang, Marc de Kamps, and Owen Johnson. Predicting pathway left shifts in hospital care using predictive process monitoring. Unpublished manuscript, School of Computer Science, University of Leeds, 2025.
- [2] NHS England. Bed availability and occupancy data – overnight, quarter 4 2025/26. <https://www.england.nhs.uk/statistics/statistical-work-areas/bed-availability-and-occupancy/>, May 2026. Published 21 May 2026. Accessed 12 August 2026.
- [3] NHS England and NHS Improvement. NHS operational planning and contracting guidance 2020/21. <https://www.england.nhs.uk/publication/nhs-operational-planning-and-contracting-guidance-2020-21/>, January 2020. Publishing Approval Reference 001300.
- [4] Nuffield Trust. Delayed discharges from hospital. <https://www.nuffieldtrust.org.uk/resource/delayed-discharges-from-hospital>, June 2026. QualityWatch indicator, last updated 4 June 2026. Length-of-stay data obtained by freedom of information request to NHS Digital.
- [5] OECD. *Health at a Glance 2025: OECD Indicators*. OECD Publishing, Paris, 2025.
- [6] Wil M. P. van der Aalst. *Process Mining: Data Science in Action*. Springer, Berlin, Heidelberg, 2nd edition, 2016.
- [7] Lerina Aversano, Martina Iammarino, Antonella Madau, Giuseppe Pirlo, and Gianfranco Semeraro. Process mining applications in healthcare: a systematic literature review. *PeerJ Computer Science*, 11:e2613, 2025.
- [8] Eric Rojas, Jorge Munoz-Gama, Marcos Sepúlveda, and Daniel Capurro. Process mining in healthcare: A literature review. *Journal of Biomedical Informatics*, 61:224–236, 2016.
- [9] Irene Teinemaa, Marlon Dumas, Marcello La Rosa, and Fabrizio Maria Maggi. Outcome-oriented predictive process monitoring: Review and benchmark. *ACM Transactions on Knowledge Discovery from Data*, 13(2):1–57, 2019.
- [10] Alistair E. W. Johnson, Lucas Bulgarelli, Lu Shen, Alvin Gayles, Ayad Shammout, Steven Horng, Tom J. Pollard, Sicheng Hao, Benjamin Moody, Brian Gow, Li-wei H. Lehman, Leo A. Celi, and Roger G. Mark. MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data*, 10(1):1, 2023.

- [11] Rui Lu, Srisuluk Kacha, Natsuda Phothikun, Atirut Supphapipat, Siwakorn Chanchalotorn, Christsanutth Siripakkaphant, Naphong Siriniran, Maethawit Chienprasert, and Kaweesak Chittawatanarat. Incidence, reasons, and mortality risk of ICU readmission in surgical patients: a systematic review and meta-analysis. *The American Journal of Surgery*, 247:116460, 2025.
- [12] Andrew A. Kramer, Thomas L. Higgins, and Jack E. Zimmerman. Intensive care unit readmissions in U.S. hospitals: patient characteristics, risk factors, and outcomes. *Critical Care Medicine*, 40(1):3–10, 2012.
- [13] Sarah Vollam, Susan Dutton, Sallie Lamb, Tatjana Petrinic, J. Duncan Young, and Peter Watkinson. Out-of-hours discharge from intensive care, in-hospital mortality and intensive care readmission rates: a systematic review and meta-analysis. *Intensive Care Medicine*, 44(7):1115–1129, 2018.
- [14] Jorge Munoz-Gama, Niels Martin, Carlos Fernandez-Llatas, Owen A. Johnson, Marcos Sepúlveda, Emmanuel Helm, Victor Galvez-Yanjari, Eric Rojas, Antonio Martinez-Millana, Davide Aloini, et al. Process mining for healthcare: Characteristics and challenges. *Journal of Biomedical Informatics*, 127:103994, 2022.
- [15] Chiara Di Francescomarino and Chiara Ghidini. Predictive process monitoring. In Wil M. P. van der Aalst and Josep Carmona, editors, *Process Mining Handbook*, volume 448 of *Lecture Notes in Business Information Processing*, pages 320–346. Springer, 2022.
- [16] Aravind Aravazhi, Berit I. Helgheim, and Petter Aadahl. Decision-making based on predictive process monitoring of patient treatment processes: A case study of emergency patients. *Advances in Operations Research*, 2023:8867057, 2023.
- [17] Toki Tahmid Inan, Mahmuda Binte Rashid Samia, Iffat Tamanna, and Muhammad Nazrul Islam. A decision support model to predict ICU readmission through data mining approach. In *Twenty-Second Pacific Asia Conference on Information Systems (PACIS)*, Japan, 2018.
- [18] Chih-Chou Chiu, Chung-Min Wu, Te-Nien Chien, Ling-Jing Kao, and Chengcheng Li. Predicting ICU readmission from electronic health records via BERTopic with long short term memory network approach. *Journal of Clinical Medicine*, 13(18):5503, 2024.
- [19] Juan C. Rojas, Kyle A. Carey, Dana P. Edelson, Laura R. Venable, Michael D. Howell, and Matthew M. Churpek. Predicting intensive care unit readmission with machine learning using electronic health record data. *Annals of the American Thoracic Society*, 15(7):846–853, 2018.

- [20] Abdulaziz Aljebreen, Allan Pang, Marc de Kamps, and Owen Johnson. Predicting unplanned hospital readmissions using outcome-oriented predictive process mining. In Andrea Delgado and Tijs Slaats, editors, *ICPM 2024 Workshops*, volume 533 of *Lecture Notes in Business Information Processing*, pages 421–433. Springer, 2025.
- [21] Y. Su, Y. Li, W. Yang, X. Luo, and L. Chen. Optimized machine learning model for predicting unplanned reoperation after rectal cancer anterior resection. *European Journal of Surgical Oncology*, 50(12):108703, 2024.
- [22] Matthew M. Ruppert, Tyler J. Loftus, Coulter Small, Han Li, Tezcan Ozrazgat-Baslanti, Jeremy Balch, Reed Holmes, Patrick J. Tighe, Gilbert R. Upchurch, Philip A. Efron, Parisa Rashidi, and Azra Bihorac. Predictive modeling for readmission to intensive care: A systematic review. *Critical Care Explorations*, 5(1):e0848, 2023.
- [23] Emanuele Koumantakis, Konstantina Remoundou, Nicoletta Colombi, Carmen Fava, Ioanna Roussaki, Alessia Visconti, and Paola Berchialla. Deep learning models for ICU readmission prediction: A systematic review and meta-analysis. *Critical Care*, 29(1):442, 2025.
- [24] Marzyeh Ghassemi, Tristan Naumann, Peter Schulam, Andrew L. Beam, Irene Y. Chen, and Rajesh Ranganath. A review of challenges and opportunities in machine learning for health. *AMIA Joint Summits on Translational Science Proceedings*, 2020:191–200, 2020. PMID: 32477638.
- [25] Michael Hagmann, Shigehiko Schamoni, and Stefan Riezler. Validity problems in clinical machine learning by indirect data labeling using consensus definitions. In *Machine Learning for Health (ML4H) Symposium, Findings Track*, New Orleans, LA, USA, December 2023.
- [26] Intensive Care Society. Levels of adult critical care second edition consensus statement. <https://ics.ac.uk/resource/levels-of-care.html>, March 2021.

Appendices

Appendix A

External Material

A.1 Systematic reviews of ICU readmission prediction

Table A.1: Systematic reviews of ICU readmission prediction: scope, synthesis instrument, and treatment of the outcome definition.

Review	Scope	Corpus size	Synthesis and findings	Treatment of outcome definition
Ruppert et al. (2023)	All model classes; 2010–2021	33 studies; model count not stated	Narrative; no meta-analysis. AUROC 0.51–0.94; homogeneous populations perform better; 6/33 externally validated	Windows range 6 hours to 30 days; 13 combined endpoints; 6 studies account for reason for readmission; devotes a recommendations section to outcome definition
Koumantakis et al. (2025)	Deep learning only; inception to March 2025	24 studies, 49 models	Meta-analysis, 11 of 24 studies. Pooled AUROC 0.79, I^2 99.96%; 2/24 externally validated, both degraded	30-day window in 16/24, same hospital stay in 5, mortality composites in 6; variation quantified and used to stratify meta-analysis subgroups

Appendix B

Care-unit mappings

B.1 Activity grouping

Table B.1: Aggregation of the 46 `careunit` values in MIMIC-IV v3.1 into the five location groups used to build the event log (subsection 3.1.3). Counts in parentheses give the units per group; the nine unmapped units are listed in Appendix B.2.

Location group	Care units
Emergency Department (2)	Emergency Department; Emergency Department Observation
ICU (8)	MICU; CVICU; MICU/SICU; SICU; TSICU; CCU; Neuro SICU; ICU
Surgery Ward (6)	Surgery; Surgery/Trauma; Cardiac Surgery; Thoracic Surgery; Surgery/Pancreatic/Biliary/Bariatric; Med/Surg/Trauma
Stepdown / Intermediate (8)	PACU; Hematology/Oncology Intermediate; Medicine/Cardiology Intermediate; Cardiology Surgery Intermediate; Neuro Intermediate; Neuro Stepdown; Surgery/Vascular/Intermediate; Surgical Intermediate
Medical Ward (13)	Medicine; Med/Surg; Medicine/Cardiology; Neurology; Hematology/Oncology; Vascular; Psychiatry; Cardiology; Oncology; Transplant; Med/Surg/GYN; Observation; Discharge Lounge

B.2 Care units outside the mapping

Table B.2: Care units present in `transfers` but matching no branch of the mapping, with transfer row counts. Of the 546,354 rows under the two unknown labels, 546,024 are discharge events.

Care unit	Transfer rows
UNKNOWN	546,196
Labor & Delivery	29,003
Obstetrics (Postpartum & Antepartum)	16,919
Medical/Surgical (Gynaecology)	10,214
Obstetrics Postpartum	8,086
Obstetrics Antepartum	3,009
Nursery	1,972
Unknown (lower case)	158
Special Care Nursery (SCN)	3

Appendix C

Pipeline reference

C.1 Source tables

Table C.1: MIMIC-IV v3.1 tables used and the attributes each supplies (subsection 3.1.1).

Table	Supplies
admissions, patients	Age band, gender, insurance, ethnicity, marital status
admissions	UHR count, admission type, admission location
omr	BMI category
diagnoses_icd	18 ICD chapter counts, Charlson score, chronic disease count
labevents	Percentage of abnormal laboratory results
prescriptions	Medication count, polypharmacy flag
transfers	The event log, both prediction points, six of the seven process features, and, via the event log, the ED flag

C.2 Process model, dominant pathway

Table C.2: The 39 independent variables, by group. One-hot encoding of the eight categorical features expands these to 76 model inputs.

Group	n	Contents
Demographic	5	Age band (eight ten-year intervals), gender, insurance (4), ethnicity (8), marital status
Clinical	27	UHR count, ED flag, admission type, BMI category, percentage of abnormal laboratory results, medication count, polypharmacy, chronic disease count, 18 ICD chapter counts, Charlson comorbidity index
Process	7	Admission location, accumulated duration, ICU and SW duration, ICU and SW stay counts, total events

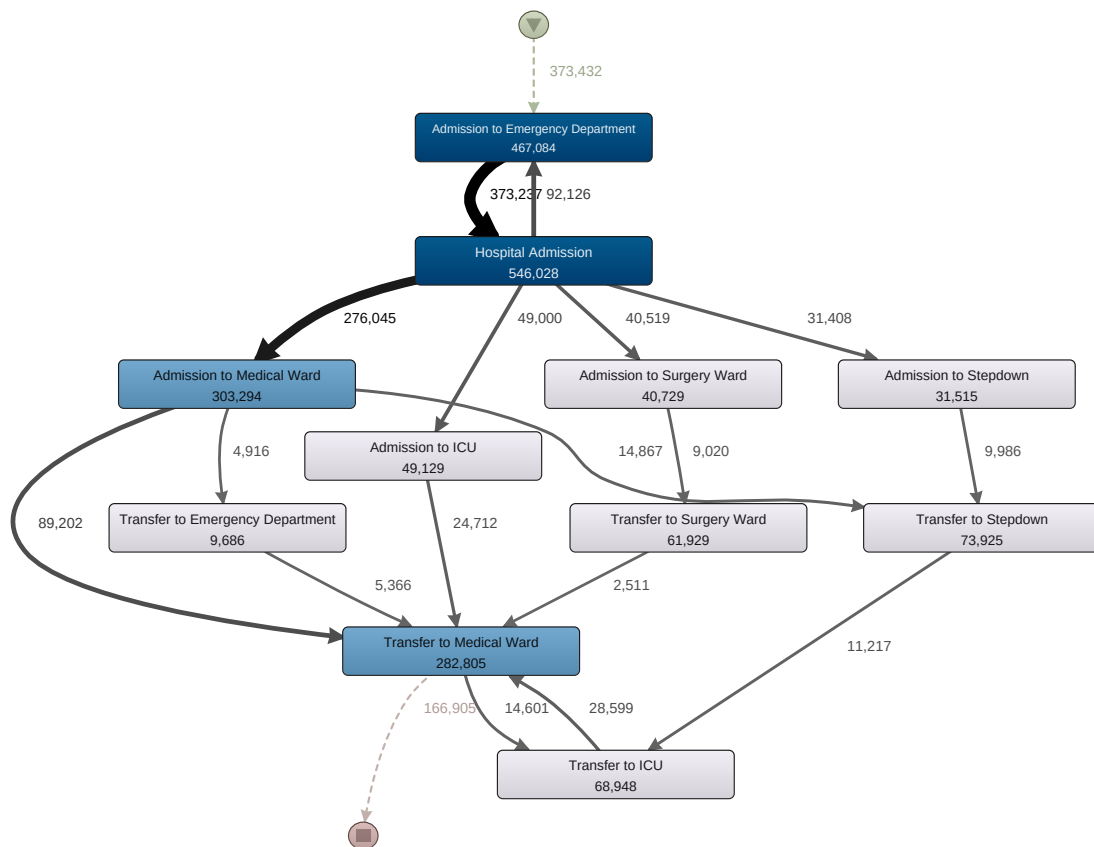


Figure C.1: The model of Figure 3.2 with the path filter at its most aggressive setting, retaining only the dominant flow: admission, emergency department, ward or intensive care, step-down, discharge. Care intensity rises and then falls, which is the pathway subsection 1.1.3 describes.

C.3 Cohort process models

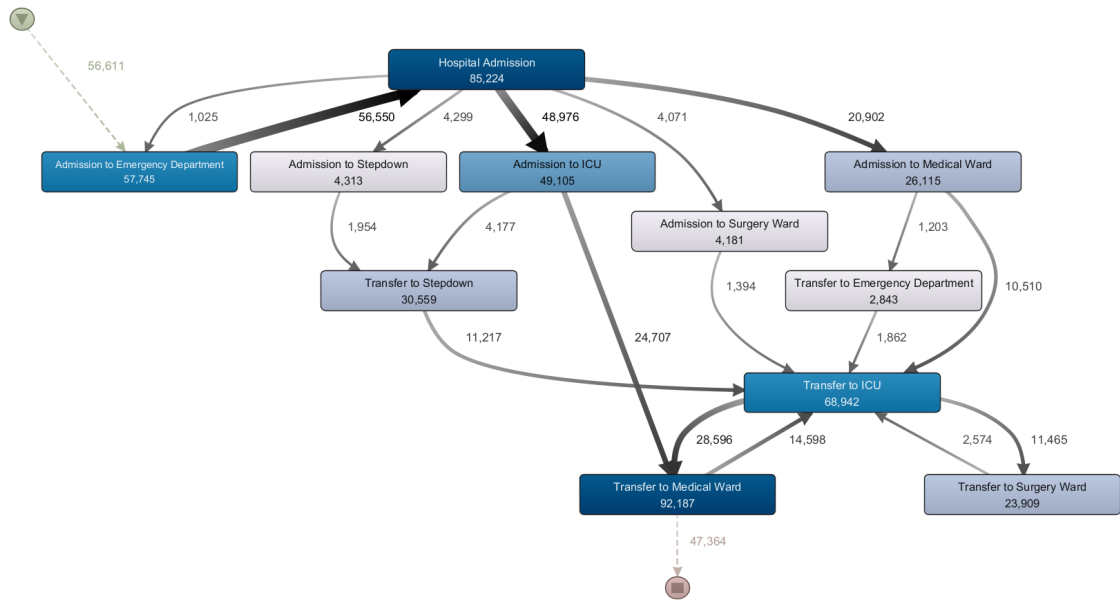


Figure C.2: Directly-follows model of the ICU cohort, 85,224 cases and 445,123 events over eleven activities. Activities 100%, paths 0%, absolute frequency.

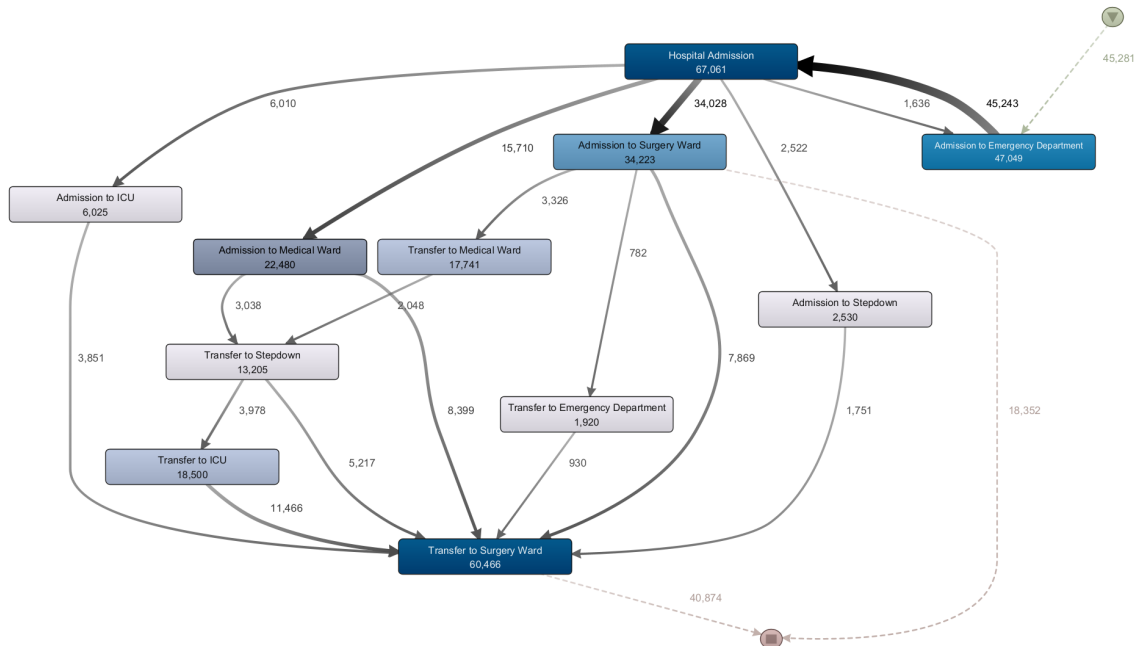


Figure C.3: Directly-follows model of the surgical ward cohort, 67,061 cases and 291,200 events over eleven activities. Same filter settings as Figure C.2.

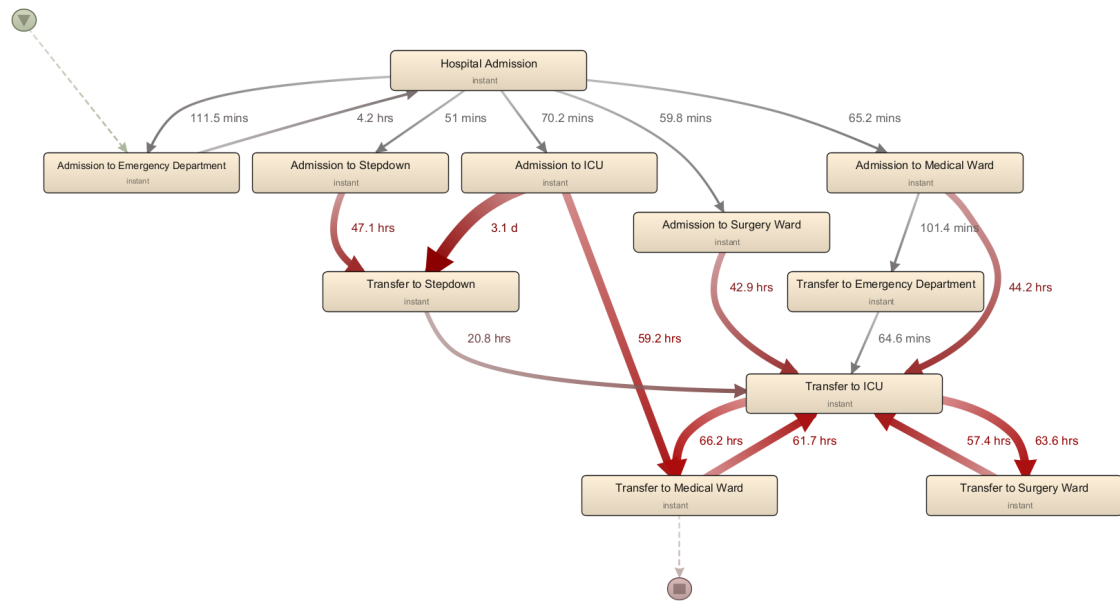


Figure C.4: The model of Figure C.2 with edges labelled by mean duration between activities.

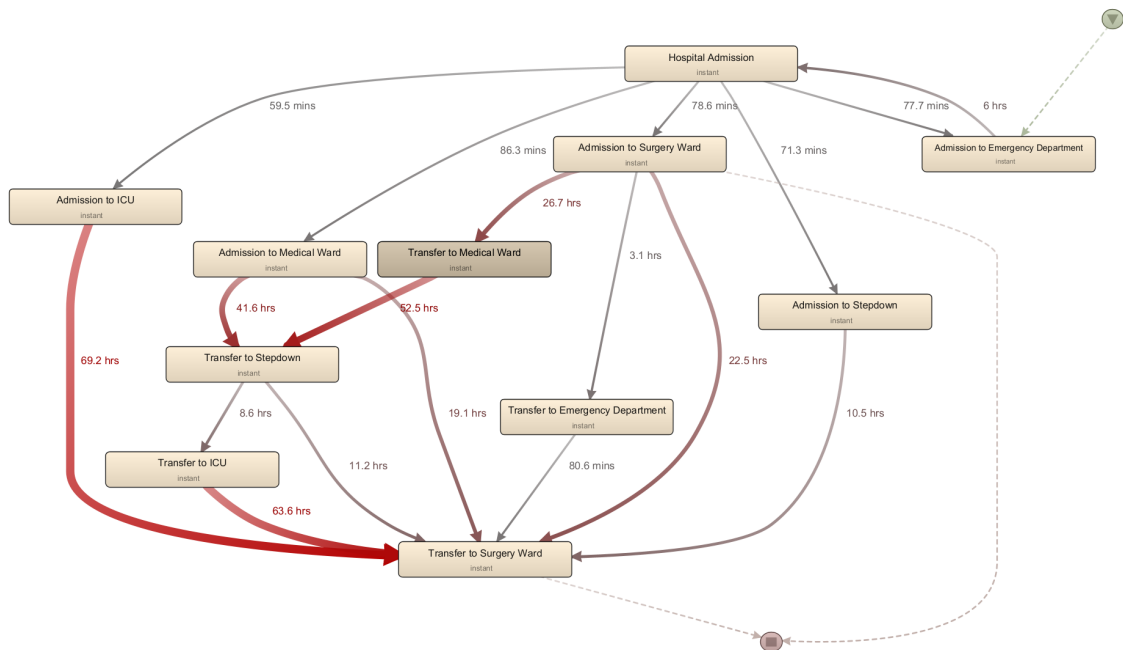


Figure C.5: The model of Figure C.3 with edges labelled by mean duration between activities.

C.4 The outcome as a process model

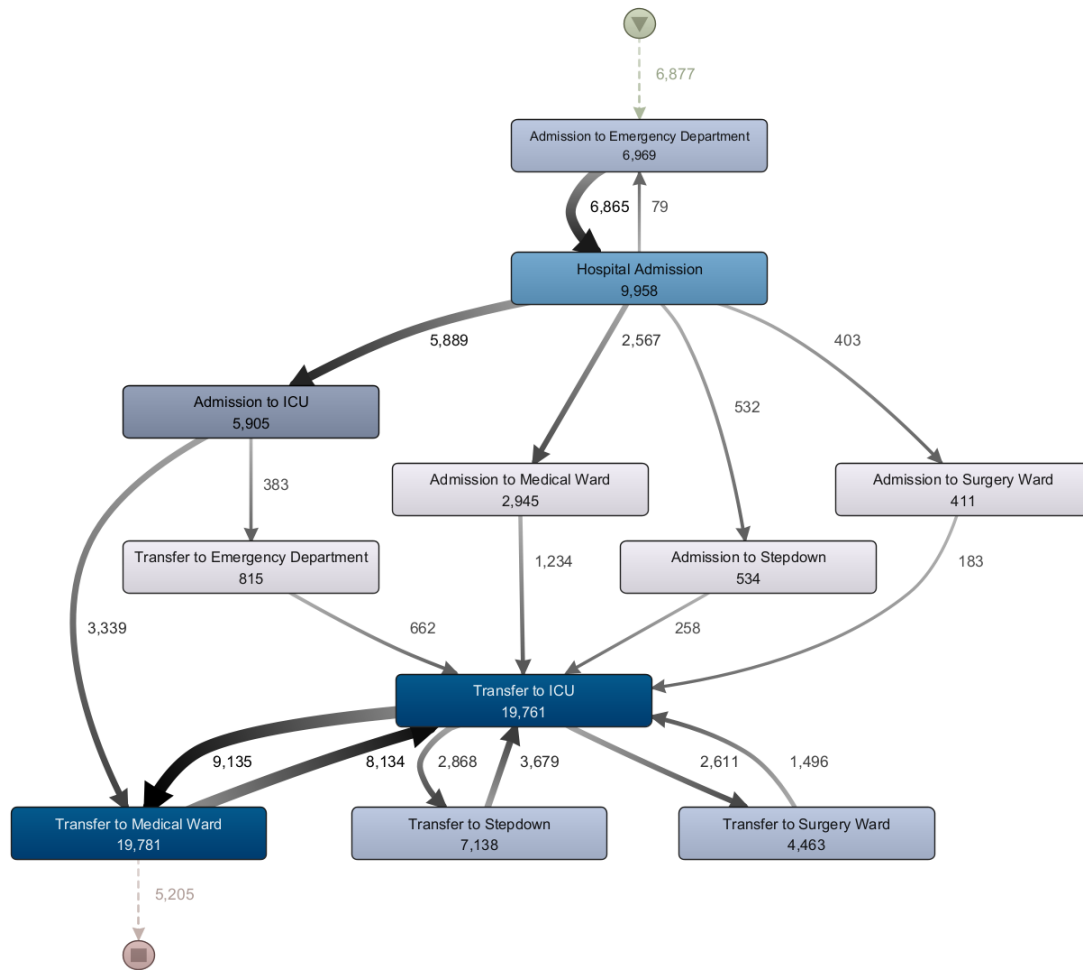


Figure C.6: ICU cohort filtered to the 9,958 admissions positive under the 1at label.

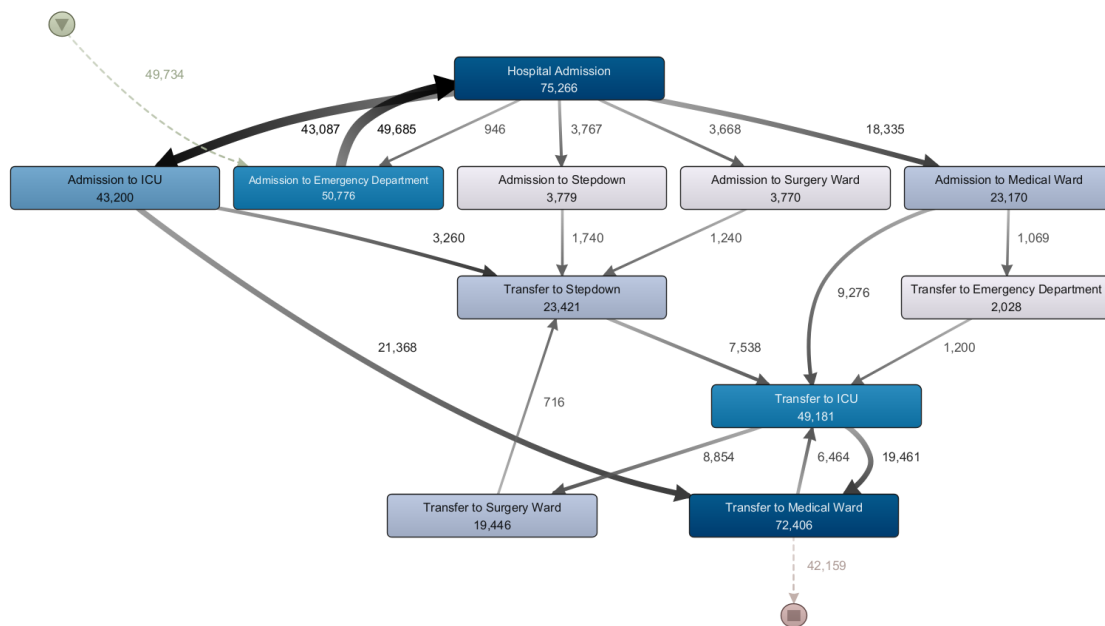


Figure C.7: The 75,266 admissions negative under `lat`. *Transfer to ICU* carries 49,181 events across these cases, 0.65 per case, against 19,761 across 9,958 in Figure C.6, or 1.98 per case. The edge is not absent from the negative class because `lat` excludes returns by patients who never left the tier.

C.5 Control-flow representations

Table C.3: The two control-flow representations of subsection 3.1.7, with their configuration and resulting dimensions in each cohort.

	Bag-of-words n-grams	Integer sequences
Configuration	Range 1–3, <code>min_df</code> = 2, vocabulary fitted on the training partition only	Vocabulary 12 (11 activities plus padding), pre-padded to length 30
ICU	280 features; matrix 322 columns	$85,224 \times 30$; mean true length 3.418, maximum 20
SW	268 features; matrix 310 columns	$73,559 \times 30$; mean true length 3.376, maximum 29

C.6 Software environments

Table C.4: Software environments (subsection 3.2.8). Classical models, the MLP, the ablations and the error analysis ran locally; the eight sequence architectures ran on Aire. Versions were read from both environments after the runs completed.

Component	Local workstation	Aire
Operating system	Windows, x86_64	Linux
Python	3.13.7	3.11.15
scikit-learn	1.7.2	1.9.0
NumPy	2.3.3	2.4.6
pandas	2.3.2	3.0.3
PyTorch	2.6.0+cu124	2.5.1+cu121
CUDA (torch build)	12.4	12.1
GPU	NVIDIA GTX 1650	NVIDIA L40S
PostgreSQL	18.2	—

C.7 Surgical ward baseline replication

Table C.5: Surgical ward baseline replication against Aljebreen et al. [1], No Process condition, AUROC (section 4.2). Values are the mean over five seeds. Δ is this project’s value less the published value.

Model	Published	lat		old	
		AUROC	Δ	AUROC	Δ
Logistic regression	0.80	0.7634	−0.0366	0.7064	−0.0936
Gradient boosting	0.82	0.7705	−0.0495	0.7134	−0.1066
Random forest	0.80	0.7540	−0.0460	0.6970	−0.1030
k -nearest neighbours	0.71	0.6408	−0.0692	0.5985	−0.1115

C.8 Single-group and incremental ablation

Table C.6: Single-group and incremental ablation over the six feature groups, `lat` label variant (section 4.5). *Alone* is the AUROC of that group by itself; *cumulative* adds groups in the order shown and *gain* is the increase on the previous row. Logistic regression throughout, fitted once per configuration, so the values are point estimates on a single test partition.

Group	Features	ICU		Surgical ward	
		Alone	Cumulative	Alone	Cumulative
Demographics	27	0.5190	0.5190	0.6096	0.6096
Clinical core	12	0.7409	0.7397	0.7610	0.7611
Chronic disease count	1	0.6087	0.7400	0.6133	0.7636
Charlson index	1	0.5845	0.7403	0.5621	0.7623
ICD chapters	18	0.6870	0.7439	0.6811	0.7634
Process	17	0.5592	0.7792	0.7715	0.8781

Appendix D

Ethical Issues Addressed

D.1 CITI Program Completion Report

COLLABORATIVE INSTITUTIONAL TRAINING INITIATIVE (CITI PROGRAM) COMPLETION REPORT - PART 1 OF 2 COURSEWORK REQUIREMENTS*

* Scores on this [Requirements Report](#) (Part 1) reflect quiz completions at the time all requirements for the course were met. The Transcript Report (Part 2) lists more recent quiz scores, including those on optional (supplemental) course elements.

• **Name:** Fenil Kalasariya (ID: 15345858)
• **Institution Affiliation:** Massachusetts Institute of Technology Affiliates (ID: 1912)
• **Institution Email:** dtjs0367@leeds.ac.uk
• **Institution Unit:** School of computer science

• **Curriculum Group:** Human Research
• **Course Learner Group:** Data or Specimens Only Research
• **Stage:** Stage 1 - Basic Course

• **Record ID:** 75264590
• **Completion Date:** 09-Feb-2026
• **Expiration Date:** 09-Feb-2029
• **Minimum Passing:** 90
• **Reported Score*:** 100

REQUIRED AND ELECTIVE MODULES ONLY	DATE COMPLETED	SCORE
Belmont Report and Its Principles (ID: 1127)	09-Feb-2026	3/3 (100%)
History and Ethics of Human Subjects Research (ID: 498)	09-Feb-2026	5/5 (100%)
Basic Institutional Review Board (IRB) Regulations and Review Process (ID: 2)	09-Feb-2026	5/5 (100%)
Records-Based Research (ID: 5)	09-Feb-2026	4/4 (100%)
Genetic Research in Human Populations (ID: 6)	09-Feb-2026	5/5 (100%)
Populations in Research Requiring Additional Considerations and/or Protections (ID: 16680)	09-Feb-2026	5/5 (100%)
Research and HIPAA Privacy Protections (ID: 14)	09-Feb-2026	5/5 (100%)
Conflicts of Interest in Human Subjects Research (ID: 17464)	09-Feb-2026	5/5 (100%)
Massachusetts Institute of Technology (ID: 1290)	09-Feb-2026	No Quiz

For this Report to be valid, the learner identified above must have had a valid affiliation with the CITI Program subscribing institution identified above or have been a paid Independent Learner.

This document was generated on 09-Feb-2026. Verify at:
www.citiprogram.org/verify/?kcd730d27-ecf1-46da-b5ea-f65e36a4bcd-75264590

Collaborative Institutional Training Initiative (CITI Program)
101 NE 3rd Avenue
Suite 320
Fort Lauderdale, FL 33301 US

Email: support@citiprogram.org
Phone: 888-529-5929
Web: <https://www.citiprogram.org>

COLLABORATIVE INSTITUTIONAL TRAINING INITIATIVE (CITI PROGRAM)
COMPLETION REPORT - PART 2 OF 2
COURSEWORK TRANSCRIPT**

** Scores on this [Transcript Report](#) (Part 2) reflect the most current quiz completions, including quizzes on optional (supplemental) elements of the course. The Requirements Report (Part 1) lists the reported scores at the time all requirements for the course were met.

• **Name:** Fenil Kalasariya (ID: 15345858)
 • **Institution Affiliation:** Massachusetts Institute of Technology Affiliates (ID: 1912)
 • **Institution Email:** dtjs0367@leeds.ac.uk
 • **Institution Unit:** School of computer science

• **Curriculum Group:** Human Research
 • **Course Learner Group:** Data or Specimens Only Research
 • **Stage:** Stage 1 - Basic Course

• **Record ID:** 75264590
 • **Current Score**:** 100

REQUIRED, ELECTIVE, AND SUPPLEMENTAL MODULES	MOST RECENT	SCORE
History and Ethics of Human Subjects Research (ID: 498)	09-Feb-2026	5/5 (100%)
Basic Institutional Review Board (IRB) Regulations and Review Process (ID: 2)	09-Feb-2026	5/5 (100%)
Belmont Report and Its Principles (ID: 1127)	09-Feb-2026	3/3 (100%)
Records-Based Research (ID: 5)	09-Feb-2026	4/4 (100%)
Genetic Research in Human Populations (ID: 6)	09-Feb-2026	5/5 (100%)
Populations in Research Requiring Additional Considerations and/or Protections (ID: 16680)	09-Feb-2026	5/5 (100%)
Research and HIPAA Privacy Protections (ID: 14)	09-Feb-2026	5/5 (100%)
Conflicts of Interest in Human Subjects Research (ID: 17464)	09-Feb-2026	5/5 (100%)
Massachusetts Institute of Technology (ID: 1290)	09-Feb-2026	No Quiz

For this Report to be valid, the learner identified above must have had a valid affiliation with the CITI Program subscribing institution identified above or have been a paid Independent Learner.

This document was generated on 09-Feb-2026. Verify at:
www.citiprogram.org/verify/2kcd730d27-ecf1-46da-b5ea-f65e36a4bcdd-75264590

Collaborative Institutional Training Initiative (CITI Program)
 101 NE 3rd Avenue
 Suite 320
 Fort Lauderdale, FL 33301 US

Email: support@citiprogram.org
 Phone: 888-529-5929
 Web: <https://www.citiprogram.org>

13/08/2026 16:13

Signed Data Use Agreement

If I am granted access to the database:

- SIGNED: **Fenil Kalasariya**

DATED: Feb. 13, 2026